ProveIt research continuation

Mixed Resolvent Products, Higher Digamma Powers, and Stieltjes Collisions

Exact identities and normalized finite parts

11 October 2026 · Version 1.0

Abstract

We develop three extensions of the current ProveIt polylogarithm and Hurwitz–Stieltjes program. First, we obtain the Fourier spectrum and Hurwitz transform of any finite product of cotangent resolvents with independent complex powers. Opposite half-planes produce additional endpoint poles; explicit full-amplitude subtraction gives a jointly holomorphic germ whose coefficients are all prescribed Stieltjes, argument, and exponent derivatives. An elementary Gamma specialization exhibits genuinely path-dependent uncompleted limits. A finite depth-two polylogarithmic formula evaluates the mixed logarithmic transform at every Stieltjes index.

Second, we prove a terminating harmonic reduction and a normalized antiderivative for every positive integer power of the digamma function. We evaluate the fifth finite-part moment explicitly, retaining one depth-four Laurent coefficient and a convergent depth-two derivative combination, with all ordinary constants determined. The proof fixes the entire Mittag–Leffler remainder and the finite asymptotic constants.

Third, we extend geometric collision subtraction to arbitrary Stieltjes indices, ordinary argument derivatives, and analytic local weights. Hermite interpolation removes the rational singular denominator before the collision limit is taken. The limit is uniform with respect to arbitrarily hierarchical gaps. The counterterms reduce to finite generalized polylogarithms; three mixed cubic examples give explicit, unequal order-zero anomalies.

We separate these proved statements from unresolved arithmetic reductions, update the status of overlapping incoming questions, and provide reproducible exact and numerical checks. No proof of the frozen Gaussian S_6 or revised S_8 conjecture is claimed.

Source baseline: ProveIt revision `9035bb0eaaf70f3398711b489733093dcc52e946`.
Prepared as an independently buildable research article with modular LaTeX sources, verification code, recorded results, and integration notes.

Contents

1 Scope, conventions, and the role of complete local subtraction 1
1.1 Relation to the current manuscript and incoming reports . . . 1
1.2 Special-function and coefficient conventions . . . 2
1.3 Coordinate finite parts . . . 2

2 Products of resolvents with independent complex powers 3
2.1 Branches and notation . . . 3
2.2 An explicit Gamma series for the full Fourier spectrum . . . 4
2.3 Hurwitz, Stieltjes, and log-Gamma transforms . . . 5
2.4 The full completion at each total-exponent resonance . . . 7
2.5 An exactly computable mixed example . . . 8
2.6 All polynomial moments and parameter derivatives . . . 9

3 Mixed logarithmic moments in finite polylogarithmic depth 10
3.1 Two and three ordinary logarithmic moments . . . 10
3.2 A depth-two transform with an arbitrary spectral order . . . 11
3.3 All Stieltjes indices and all argument derivatives . . . 12
3.4 A special-point identity involving a harmonic Laurent coefficient . . . 13
3.5 A generating identity for every mixed logarithmic moment . . . 14
3.6 Derivatives and primitives in two resolvent parameters . . . 14

4 A finite harmonic reduction for every digamma power 15
4.1 The coefficients and the uniform theorem . . . 15
4.2 The entire remainder and the normalized primitive . . . 16
4.3 Proof of the finite reduction . . . 17

5 The fifth-power identity and its additional spectral coordinate 18
5.1 Explicit principal parts and every finite constant . . . 19
5.2 The moving-order identity and the proof of the bridge . . . 21
5.3 The highest-depth coefficient and a finite cancellation . . . 21
5.4 Verification and the exact scope of the result . . . 22

6 Polylogarithmic counterterms for higher Stieltjes collisions 22
6.1 Local singular polynomials and the finite-part convention . . . 23
6.2 A finite Hermite construction . . . 23
6.3 One logarithmic index: an explicit dilogarithm kernel . . . 26
6.4 Three new mixed cubic collision constants . . . 27
6.5 An ordinary integral for the remaining merged constant . . . 28

7 Audit, integration corrections, and further research 29
7.1 What the audit changes . . . 29
7.2 Exact corrections to tempting extrapolations . . . 30
7.3 Questions opened by the new results . . . 30
7.4 Suggested order of work . . . 32

8 Verification, reproducibility, and the evidence boundary 33
8.1 Exact calculations and independent numerical methods . . . 33
8.2 Independent geometric-collision checks . . . 33
8.3 Companion files and replay . . . 34

1 Scope, conventions, and the role of complete local subtraction

1.1 Relation to the current manuscript and incoming reports

The source baseline is the repository revision `9035bb0eaaf70f3398711b489733093dcc52e946`, whose manuscript is organized into four volumes and whose current incoming directory contains seventeen ZIP reports [1]. The work here follows the Hurwitz–Stieltjes and harmonic-identity spine. The source review included the manuscript's structure and status records, the relevant finite-part chapters, and the incoming reports bearing on the questions below. It is a targeted mathematical audit, not a claim that every theorem in the four volumes and all incoming archives has been re-proved.

There are already substantial results at this baseline. The cubic and quartic digamma bridges, single-resolvent complex powers at nonpositive resonances, all-degree formal Tornheim relation spaces, and the Stieltjes-index-zero collision theorem are prior work [4, 3, 2, 5]. Several questions in older incoming reports have consequently become stale. We preserve that distinction throughout: a theorem that was already proved in a newer report is not presented as a new result here.

The concrete contributions and their limits are as follows.

Target	Result in this article
Several resolvents with independent exponents	Theorems 2.1, 2.3, and 2.5 give the complete Fourier spectrum, the Hurwitz transform, and the local completion for any finite number of nonreal parameters. Opposite orientations are included. This answers the transformation and normalization portions of Question 10 in [2].
Mixed logarithmic products	Theorem 3.3 gives a finite expression of depth at most two for two opposite logarithmic resolvents, at every Stieltjes index and every ordinary argument-derivative order. It supplies a specific instance of the mixed-product question in [2].
The fifth digamma moment	Theorem 5.1 resolves Question 1 of [2], with all finite corrections and a convergent polygamma representation. It also supplies the first new case requested by Question F3 of [3].
Arbitrary digamma powers	Theorem 4.1 gives a terminating reduction to ordinary convergent multiple zeta values and specified first Laurent coefficients. Corollary 4.3 fixes an antiderivative for every power. No claim of an arithmetically minimal output basis is made.
Higher-index geometric collisions	Theorem 6.2 proves the total-collision limit for arbitrary Stieltjes indices and ordinary derivatives, with an analytic local weight and no restriction on gap ratios. This resolves the higher-index extension posed in [5] at the level of continuity after complete subtraction.
Gaussian period conjectures	The frozen S_6 and revised S_8 remain open. Neither a new period certificate nor a counterexample is obtained. The earlier rejected S_8 vector and the stated formal relation obstructions retain their separate established status [1, 6].

The word *new* refers here to progress relative to this pinned source corpus. Classical beta integrals, Gamma expansions, harmonic products, and multiple-zeta dualities are cited as such. The external literature check was focused; it does not establish global priority for every consequence or specialization.

1.2 Special-function and coefficient conventions

The generalized Stieltjes functions are normalized by

$$\zeta(1+u,x)=\frac{1}{u}+\sum_{m\ge 0}\frac{(-1)^m}{m!}\gamma_m(x)u^m,\qquad \gamma_m=\gamma_m(1),\quad \gamma=\gamma_0. \tag{1.1}$$

Thus $\gamma_0(x)=-\psi(x)$, where $\psi=\Gamma'/\Gamma$. Equivalently,

$$\zeta(1-s,x)+\frac{1}{s}=\sum_{m\ge 0}\gamma_m(x)\frac{s^m}{m!}.$$

A superscript (p) on γ_m or ψ means an ordinary derivative in its argument. A prime on a zeta function of a single variable denotes an order derivative unless an argument is explicitly specified. The standard Hurwitz recurrence and differentiation formulas are [8, 10]

$$\gamma_m(x)=\frac{\log^m x}{x}+\gamma_m(1+x), \tag{1.2}$$

$$\partial_x^p\zeta(s,x)=(-1)^p(s)_p\zeta(s+p,x). \tag{1.3}$$

Here $(s)_p=s(s+1)\cdots(s+p-1)$ is the rising factorial. The symbol $[u^j]F$ means a Taylor or Laurent coefficient, including no factorial. A derivative is recovered by multiplying a Taylor coefficient by the appropriate factorial. This distinction matters at every harmonic Laurent coefficient used below.

Our multiple zeta convention is strict and decreasing:

$$Z_d(s_1,\dots,s_d)=\sum_{n_1>\cdots>n_d\ge 1}n_1^{-s_1}\cdots n_d^{-s_d},$$

in its convergence region, with meromorphic continuation where stated. The notation $\zeta(s,t)$ in the polylogarithmic section denotes the same depth-two function. For a nonnegative integer N, $H_N^{(r)}=\sum_{n=1}^N n^{-r}$; multiple harmonic words use the same strict ordering. No weakly ordered sum is silently substituted for a strict one.

1.3 Coordinate finite parts

All endpoint finite parts are taken in the original integration coordinate. For an integrand with a finite power–log singular expansion, integrate after removing a small endpoint interval, expand the result in the cutoff, and retain its constant term. At two distinct endpoints the cutoffs are independent. In particular,

$$\operatorname{FP}\int_0^1 x^{\alpha-1}\log^j x\,\mathrm{d}x=\frac{(-1)^j j!}{\alpha^{j+1}}\qquad(\alpha\neq 0), \tag{1.4}$$

$$\operatorname{FP}\int_0^1 x^{-1}\log^j x\,\mathrm{d}x=0\qquad(j\ge 0). \tag{1.5}$$

The first equality follows by meromorphic continuation from $\Re\alpha>0$, or directly from the cutoff primitive. At $\alpha=0$ the second equality is the coordinate constant; it is not a value obtained by inserting zero into the first formula.

For periodic Stieltjes products, the local singularity occurs on the right of each seam. We remove a right interval of ambient length η at every singular point and take independent constant terms. Ordinary derivatives on the open intervals are used throughout. Differentiating a periodic distribution that has already been extended across a seam is a different operation and can carry additional distributional contacts. Section 6 specifies its convention explicitly.

For fixed distinct singular points, independent finite parts are linear and commute with fixed parameter derivatives once enough local monomials have been subtracted uniformly. This elementary fact is the common analytic step behind the spectral and geometric constructions. It does not authorize taking an arbitrary path through intersecting parameter poles.

Lemma 1.1 (Full-amplitude completion)**.** *Let a parameter-dependent integrand near an endpoint have a finite expansion whose singular terms are $A_j(\mathbf{t})x^{\alpha_j(\mathbf{t})-1}$, with holomorphic amplitudes and affine exponents, none identically zero. Suppose the subtracted remainder and each fixed parameter derivative are uniformly integrable near $\mathbf{t} = 0$. For every j such that $\alpha_j(0) = 0$, subtract $A_j(\mathbf{t})/\alpha_j(\mathbf{t})$ from its meromorphically continued integral. The resulting local germ is holomorphic, and its Taylor coefficients are exactly the coordinate finite parts of the corresponding derivatives of the integrand. Power–log monomials follow by differentiating in an auxiliary exponent.*

Proof. On a fixed endpoint interval $(0, \rho)$, the integrated monomial is

$$A_j(\mathbf{t})\frac{\rho^{\alpha_j(\mathbf{t})}}{\alpha_j(\mathbf{t})} = \frac{A_j(\mathbf{t})}{\alpha_j(\mathbf{t})} + A_j(\mathbf{t})\frac{\rho^{\alpha_j(\mathbf{t})} - 1}{\alpha_j(\mathbf{t})}.$$

The second term is entire in α_j and generates the cutoff constants of all its logarithmic derivatives. Nonresonant monomials are already holomorphic at the base point. Uniform integrability justifies coefficient extraction from the remainder. The artificial splitting point cancels against the rest of the integral. □

The amplitude in this lemma is the complete function $A_j(\mathbf{t})$, not merely its restriction to the polar hyperplane. The latter specifies a residue but omits regular coefficients that contribute to the prescribed finite part. The two-orientation example in Section 2 makes this distinction exact: different uncompleted parameter paths give different finite limits for the same specialized function.

2 Products of resolvents with independent complex powers

Question 10 of the incoming *Quartic Tails and Complex Resolvents* report asks for products of several nonreal cotangent resolvents, including parameters in opposite half-planes [2]. We answer its analytic and transformation parts for any finite number of factors. The answer has two distinct features: every Fourier coefficient has a geometrically convergent Gamma-series representation, and the complete endpoint subtraction depends on a signed sum of exponents as well as on their total. The latter dependence produces genuine extra poles when both half-planes occur.

2.1 Branches and notation

Let $z_1, \dots, z_d \in \mathbb{C} \setminus \mathbb{R}$ and $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_d) \in \mathbb{C}^d$. Repeated values of the z_j are allowed. Define

$$\begin{aligned} K(x) &= \pi\cot(\pi x), & R_z(x) &= (K(x) - z)^{-1}, & & \\ \epsilon_j &= \operatorname{sgn}\Im z_j, & q_j &= \frac{z_j - \epsilon_j\pi\mathrm{i}}{z_j + \epsilon_j\pi\mathrm{i}}, & C_j &= -\frac{1}{z_j + \epsilon_j\pi\mathrm{i}}, \\ a &= \sum_{\epsilon_j=1} \lambda_j, & b &= \sum_{\epsilon_j=-1} \lambda_j, & \Lambda &= a + b, \qquad E = a - b. \end{aligned} \tag{2.1}$$

In particular $|q_j| < 1$. In every occurrence of $C_j^{\lambda_j}$ use

$$\operatorname{Log} C_j = \operatorname{Log}(1 - q_j) - \log(2\pi) + \epsilon_j\pi\mathrm{i}/2,$$

where $\operatorname{Log}(1 - q_j)$ is holomorphic on the disk and vanishes at $q_j = 0$. The continuous logarithm of R_z on $(0, 1)$ is fixed by $\operatorname{Log} R_z(x) - \log x \to 0$ as $x \downarrow 0$. For $\epsilon = \operatorname{sgn}\Im z$, its imaginary part lies between 0 and $\epsilon\pi$. Powers in

$$P_{\boldsymbol{\lambda}}(x) = \prod_{j=1}^{d} R_{z_j}(x)^{\lambda_j} \tag{2.2}$$

use these logarithms. Put

$$B(\boldsymbol{\lambda},\mathbf{q})=\prod_j(1-q_j)^{\lambda_j},\qquad V(x;\boldsymbol{\lambda})=\prod_j\bigl(\pi x\cot\pi x-z_jx\bigr)^{-\lambda_j}.$$

The powers defining B use the holomorphic logarithms on $|q_j|<1$. The powers defining V use their analytic logarithms at $x=0$, where $V(0)=1$. Direct algebra gives both the global expression

$$P_{\boldsymbol{\lambda}}(x)=\pi^{-\Lambda}B\,\sin^{\Lambda}(\pi x)e^{\pi\mathrm{i}Ex}\prod_{\epsilon_j=1}(1-q_je^{2\pi\mathrm{i}x})^{-\lambda_j}\prod_{\epsilon_j=-1}(1-q_je^{-2\pi\mathrm{i}x})^{-\lambda_j},\tag{2.3}$$

and the two endpoint expansions

$$P_{\boldsymbol{\lambda}}(x)=x^{\Lambda}V(x;\boldsymbol{\lambda}),\qquad P_{\boldsymbol{\lambda}}(1-y)=e^{\pi\mathrm{i}E}y^{\Lambda}V(-y;\boldsymbol{\lambda}).\tag{2.4}$$

The leading coefficient at zero is exactly one. This normalization will fix every contact term below.

2.2 An explicit Gamma series for the full Fourier spectrum

Define the exponentially decreasing coefficient sequences

$$U_n=[w^n]\prod_{\epsilon_j=1}(1-q_jw)^{-\lambda_j},\qquad W_n=[w^n]\prod_{\epsilon_j=-1}(1-q_jw)^{-\lambda_j}.\tag{2.5}$$

An empty product has coefficient 1 at $n=0$ and 0 otherwise. Every coefficient is a finite sum; for example

$$U_n=\sum_{\substack{(n_j)\ge0\\ \sum_{\epsilon_j=1}n_j=n}}\prod_{\epsilon_j=1}\frac{(\lambda_j)_{n_j}q_j^{n_j}}{n_j!}.$$

For $k\in\mathbb{Z}$ set

$$G_k(a,b)=\frac{e^{\pi\mathrm{i}(a-b)/2}(-1)^k\Gamma(1+a+b)}{(2\pi)^{a+b}\Gamma(1+a-k)\Gamma(1+b+k)}.\tag{2.6}$$

Reciprocal Gamma factors are read as entire functions. Restrictions through an intersection of a numerator pole and a reciprocal-Gamma zero are made as meromorphic restrictions, never as an informal $0\cdot\infty$.

Theorem 2.1 (Fourier coefficients of any finite product)**.** *For $\Re\Lambda>-1$, the ordinary Fourier coefficients of (2.2) are*

$$c_k(\boldsymbol{\lambda},\mathbf{z}):=\int_0^1P_{\boldsymbol{\lambda}}(x)e^{-2\pi\mathrm{i}kx}\,\mathrm{d}x=B\sum_{n,m\ge0}U_nW_mG_{k-n+m}(a,b).\tag{2.7}$$

The double series converges normally, geometrically in the two summation indices on compact subsets of $|q_j|<1$, away from its possible parameter poles. It supplies a joint meromorphic continuation with possible poles only on $\Lambda=-1,-2,\dots$. Its mean is c_0.

If every z_j is in the same half-plane, then

$$c_0=\prod_jC_j^{\lambda_j}.\tag{2.8}$$

For mixed half-planes, (2.7) gives an explicit two-sided spectrum and (2.8) generally fails.

Proof. The classical Fourier beta integral [7, Eq. 5.12.6] reads

$$\int_0^1 \sin^{\Lambda}(\pi x) e^{\pi \mathrm{i} \nu x}\, \mathrm{d}x = \frac{e^{\pi \mathrm{i}\nu/2}\Gamma(1+\Lambda)}{2^{\Lambda}\Gamma(1+(\Lambda+\nu)/2)\Gamma(1+(\Lambda-\nu)/2)}, \qquad \Re\Lambda > -1.$$

Substitution $\nu = a - b - 2k$ gives G_k. Expanding only the analytic factors in (2.3), whose series converge uniformly on the circle, and integrating against the integrable endpoint factor proves (2.7).

Choose a radius $\rho > 1$ with $\rho \max_j |q_j| < 1$. Cauchy's estimate bounds U_n, W_n by compact-set constants times ρ^{-n} and ρ^{-m}. Gamma reflection and the Gamma-ratio asymptotic show that G_k has at most polynomial growth in the integer k, uniformly on compact parameter sets avoiding $\Lambda = -1, -2, \ldots$. Consequently the series and all fixed parameter derivatives converge normally there. Formula (2.6) proves the stated meromorphy.

In a common half-plane put $w = e^{2\epsilon\pi \mathrm{i} x}$. The kernel is

$$\left(\prod_j C_j^{\lambda_j}\right)(1-w)^{\Lambda}\prod_j (1-q_j w)^{-\lambda_j}.$$

Its holomorphic disk function has constant coefficient $\prod C_j^{\lambda_j}$. Abel limits are integrable when $\Re\Lambda > -1$, by the combined endpoint exponent, and give (2.8). This argument uses the total exponent, not separate inequalities on individual exponents. □

Remark 2.2 (A finite hypergeometric specialization). With one factor at $z_+ = \pi\mathrm{i}$ of exponent a and one lower-half-plane factor of exponent b and disk coordinate r, (2.7) gives

$$c_0 = (1-r)^b G_0(a,b)\, {}_2F_1\left(\begin{matrix}-a, b\\ 1+b\end{matrix}; r\right). \tag{2.9}$$

This follows from $G_m/G_0 = (-a)_m/(1+b)_m$ in the initial nonexceptional domain. At exceptional parameters the full expression is continued as a unit. Thus even one opposite orientation changes the constant-mode formula from a product of constants to a hypergeometric function. Formula (2.7), rather than a name for a new function, specifies the general answer by elementary finite coefficients and Gamma values.

2.3 Hurwitz, Stieltjes, and log-Gamma transforms

Let $p \geq 0$ be an integer and put

$$A_p(s) = \prod_{j=1}^{p}(s-j), \quad A_0(s) = 1, \qquad \sigma_+ = -2\pi\mathrm{i}, \quad \sigma_- = 2\pi\mathrm{i}, \quad \operatorname{Log}\sigma_{\pm} = \log(2\pi) \mp \pi\mathrm{i}/2.$$

Define, initially in an absolute-convergence region,

$$D_+(u) = \sum_{n\geq 1}\frac{c_n}{n^u}, \qquad D_-(u) = \sum_{n\geq 1}\frac{c_{-n}}{n^u}. \tag{2.10}$$

The definitions retain all coefficients, including their normalization B.

Theorem 2.3 (The joint spectral identity)**.** *In the ordinary-integral domain*

$$\Re\Lambda > -1, \qquad \Re(s+\Lambda) > p,$$

away from the Hurwitz pole for $p = 0$, analytic continuation from an absolutely convergent Fourier region gives

$$\begin{aligned} J_p(s,\boldsymbol{\lambda}) &:= \int_0^1 \partial_x^p \zeta(1-s,x) P_{\boldsymbol{\lambda}}(x)\, \mathrm{d}x \\ &= \Gamma(s)\left\{\sigma_+^{p-s} D_+(s-p) + \sigma_-^{p-s} D_-(s-p)\right\}. \end{aligned} \tag{2.11}$$

The completed Stieltjes generator is

$$T_p(s,\boldsymbol{\lambda}) = J_p(s,\boldsymbol{\lambda}) + \mathbf{1}_{p=0}\frac{c_0}{s}. \tag{2.12}$$

It is jointly meromorphic. Its only possible polar hypersurfaces are

$$s+\Lambda = p-j \quad (j\geq 0), \qquad \Lambda = -1-j \quad (j\geq 0). \tag{2.13}$$

For $\Re\Lambda > p$,

$$\int_0^1 \gamma_m^{(p)}(x) P_{\boldsymbol{\lambda}}(x)\,\mathrm{d}x = m![s^m]T_p(s,\boldsymbol{\lambda}). \tag{2.14}$$

Proof. In an initial region $\Re s > p+1$, $\Re\Lambda > -1$, Fourier pairing with the classical Hurwitz expansion [8, Eq. 25.11.9] gives zero for the constant mode and

$$\int_0^1 \partial_x^p \zeta(1-s,x) e^{\pm 2\pi i n x}\,\mathrm{d}x = \Gamma(s)\sigma_\pm^{p-s} n^{p-s}.$$

Here the differentiated Hurwitz series converges absolutely and uniformly, and $P_{\boldsymbol{\lambda}}$ belongs to $L^1(0,1)$. Thus termwise integration is justified directly by its uniform bound; no regularity assertion for the endpoint values of P is required. This proves (2.11) in that region.

Use the exact holomorphic splitting

$$\partial_x^p\left(\zeta(1-s,x)+\frac{1}{s}\right) = A_p(s)x^{s-p-1} + h_p(s,x), \qquad h_p(s,x) = \partial_x^p\left(\zeta(1-s,1+x)+\frac{1}{s}\right). \tag{2.15}$$

It proves normal integrability on the stated ordinary domain and gives analytic continuation of the Fourier identity to that domain. At the two endpoints, the monomial exponents are respectively $s+\Lambda-p-1+j$ and $\Lambda+j$. Subtracting any required finite number of such monomials leaves a normally convergent integral. Their integration denominators give exactly (2.13). The pole of the uncompleted Hurwitz factor is $-c_0/s$ for $p=0$; (2.12) removes it before Stieltjes coefficient extraction. The convention $\zeta(1-s,x)+s^{-1} = \sum_{m\geq 0}\gamma_m(x)s^m/m!$ then gives (2.14). □

The apparent coordinate poles of separate Dirichlet or Gamma factors in (2.11) may cancel. Equation (2.13) concerns the complete function, as proved by its endpoint integrals.

Corollary 2.4 (An ordinary log-Gamma identity)**.** *For* $\Re\Lambda > -1$,

$$\int_0^1 \log\Gamma(x) P_{\boldsymbol{\lambda}}(x)\,\mathrm{d}x = \frac{\log(2\pi)}{2}c_0 + \sum_{\epsilon\in\{+,-\}} \frac{(\gamma+\operatorname{Log}\sigma_\epsilon)D_\epsilon(1) - D_\epsilon'(1)}{\sigma_\epsilon}. \tag{2.16}$$

The prime differentiates the Dirichlet order. This is an ordinary integral; its logarithmic endpoint is integrable under the stated condition. The formula follows by differentiating (2.11) *at* $s=1$ *and using* $\zeta'(0,x) = \log\Gamma(x) - \frac{1}{2}\log(2\pi)$.

For a concrete ordinary evaluation, take the special points $z_\pm = \pm\pi\mathrm{i}$ with exponents $2,1$. Their Fourier coefficients are $\mathrm{i}(-1,3,-3,1)/(8\pi^3)$ at modes $-1,0,1,2$, respectively. Consequently (2.16) reduces to

$$\int_0^1 \log\Gamma(x)\,\frac{\sin^3(\pi x)}{\pi^3}e^{\pi\mathrm{i}x}\,\mathrm{d}x = \frac{3\{\gamma+\log(2\pi)\}-\log 2}{32\pi^4} + \frac{\mathrm{i}\{12\log(2\pi)-7\}}{64\pi^3}. \tag{2.17}$$

2.4 The full completion at each total-exponent resonance

The preceding polar arrangement is completed explicitly as follows. Set

$$v_j(\boldsymbol{\lambda}) = [x^j]V(x;\boldsymbol{\lambda}), \qquad b_j(s,\boldsymbol{\lambda}) = [x^j]h_p(s,x)V(x;\boldsymbol{\lambda}). \tag{2.18}$$

These are finite, effectively computable expressions. For instance

$$v_0 = 1, \qquad v_1 = \sum_j \lambda_j z_j, \qquad v_2 = \frac{1}{2}\left(\sum_j \lambda_j z_j\right)^2 + \frac{1}{2}\sum_j \lambda_j z_j^2 + \frac{\pi^2}{3}\Lambda.$$

For $p + j > 0$, the coefficient of x^j in h_p is

$$\frac{(-1)^{p+j}(1-s)_{p+j}}{j!}\zeta(1-s+p+j),$$

and at $p = j = 0$ it is $\zeta(1-s) + 1/s$. Thus all required local derivatives use only ordinary zeta derivatives and Stieltjes constants.

Fix $\boldsymbol{\lambda}^0$ with $L = \sum_j \lambda_j^0 \in \mathbb{Z}$, write $\boldsymbol{\lambda} = \boldsymbol{\lambda}^0 + \boldsymbol{\eta}$ and $\delta = \sum_j \eta_j$. Define

$$\begin{aligned} H_{p,\boldsymbol{\lambda}^0}(s,\boldsymbol{\eta}) = T_p(s,\boldsymbol{\lambda}^0+\boldsymbol{\eta}) &- \mathbf{1}_{L\le p}\frac{A_p(s)v_{p-L}(\boldsymbol{\lambda}^0+\boldsymbol{\eta})}{s+\delta} \\ &- \mathbf{1}_{L\le -1}\frac{\left\{1+(-1)^{-L-1}e^{\pi \mathrm{i} E(\boldsymbol{\lambda}^0+\boldsymbol{\eta})}\right\}b_{-L-1}(s,\boldsymbol{\lambda}^0+\boldsymbol{\eta})}{\delta}. \end{aligned} \tag{2.19}$$

False-indicator terms are omitted, not evaluated.

Theorem 2.5 (All Stieltjes, argument, and exponent derivatives)**.** *The function* (2.19) *is holomorphic near* $(s,\boldsymbol{\eta}) = (0,0)$*. For* $m \ge 0$ *and a multi-index* $\mathbf{k} \ge 0$*,*

$$\begin{aligned} \mathrm{FP}_x \int_0^1 \gamma_m^{(p)}(x)\prod_j R_{z_j}(x)^{\lambda_j^0}(\operatorname{Log} R_{z_j}(x))^{k_j}\,\mathrm{d}x \\ = m!\left(\prod_j k_j!\right)[s^m\boldsymbol{\eta}^{\mathbf{k}}]H_{p,\boldsymbol{\lambda}^0}(s,\boldsymbol{\eta}). \end{aligned} \tag{2.20}$$

The finite part uses the original endpoint coordinates x *and* $1-x$*. The individual exponents* λ_j^0 *need not be integers; only their total is integral in this statement. If the total is not integral, the corresponding local germ of* T_p *itself gives all the finite parts.*

Proof. At zero, (2.15) and (2.4) give the exact families

$$A_p(s)v_j(\boldsymbol{\lambda})x^{s+\Lambda-p-1+j}, \qquad b_j(s,\boldsymbol{\lambda})x^{\Lambda+j}.$$

At the other endpoint the smooth Hurwitz factor is $h_p(s,-y)$. Its product with the kernel has coefficient $e^{\pi \mathrm{i} E}(-1)^j b_j(s,\boldsymbol{\lambda})$ at $y^{\Lambda+j}$. The only exponents whose integrated denominators vanish at the base point are $j = p - L$ and $j = -L-1$. Their complete amplitudes give the two subtraction terms in (2.19).

For rigor, choose a fixed small endpoint interval $[0,\rho]$, and subtract sufficiently many of the displayed monomials that the remainders and their fixed derivatives are integrable uniformly on a parameter polydisk. Every nonresonant integrated monomial is holomorphic there. For each resonant exponent $\alpha - 1$ use

$$\frac{\rho^\alpha}{\alpha} = \frac{1}{\alpha} + \frac{\rho^\alpha - 1}{\alpha}.$$

The second term is entire; its Taylor coefficients are exactly the constant terms of the logarithmic cutoff integrals. Subtracting the first term with its full parameter-dependent amplitude therefore makes the remainder holomorphic and proves (2.20). Contributions depending on the artificial splitting point cancel between the local and remaining integrals. The same argument proves the nonintegral case, where no denominator is resonant at the base point. □

Corollary 2.6 (Why opposite half-planes produce an extra pole)**.** *On the hyperplane* $\Lambda = -M$, $M \geq 1$, *the regular-endpoint residue in* (2.19) *is*

$$(1 - e^{-2\pi \mathrm{i} b})\, b_{M-1}(s, \boldsymbol{\lambda}). \tag{2.21}$$

If all factors lie in the same half-plane this vanishes identically on the hyperplane, so its quotient is removable. With both orientations present it is generically nonzero. Its vanishing at an isolated integral parameter vector does not make that multivariable quotient removable.

Proof. On $\Lambda = -M$, $E = \Lambda - 2b = -M - 2b$, and hence $1 + (-1)^{M-1} e^{\pi \mathrm{i} E} = 1 - e^{-2\pi \mathrm{i} b}$. In a single orientation $b = 0$ or $b = \Lambda$. With two orientations, b varies independently along the hyperplane. The local coefficient b_{M-1} may introduce further cancellations; generic nonvanishing already occurs for $M = 1$ and the constant test function. □

2.5 An exactly computable mixed example

For $z_+ = \pi \mathrm{i}$, $z_- = -\pi \mathrm{i}$, write their exponents as α, β. The kernel and ordinary mean simplify to

$$P_{\alpha,\beta}(x) = \left(\frac{\sin \pi x}{\pi}\right)^{\alpha+\beta} e^{\pi \mathrm{i}(\alpha-\beta)x}, \tag{2.22}$$

$$M(\alpha, \beta) = \frac{e^{\pi \mathrm{i}(\alpha-\beta)/2} \Gamma(1 + \alpha + \beta)}{(2\pi)^{\alpha+\beta} \Gamma(1 + \alpha) \Gamma(1 + \beta)}. \tag{2.23}$$

For any $c, d \in \mathbb{C}$ with $c + d \neq 0$, continuation along $\alpha = -1 + ct$, $\beta = dt$ yields

$$\lim_{t \to 0} M(-1 + ct, dt) = -2\pi \mathrm{i} \frac{c}{c + d}. \tag{2.24}$$

The fixed-coordinate finite part of the specialized integrand is instead

$$\mathrm{FP}_x \int_0^1 (K(x) - \pi \mathrm{i})\, \mathrm{d}x = -\pi \mathrm{i}. \tag{2.25}$$

The exact correction is the full regular-endpoint quotient

$$\frac{1 + e^{\pi \mathrm{i}(\alpha-\beta)}}{1 + \alpha + \beta}.$$

Its limit on the same ray is $-\pi \mathrm{i}(c - d)/(c + d)$; subtracting it from (2.24) gives (2.25) for every ray. Thus the rays $(c, d) = (1, 0), (1, 1), (0, 1)$ yield continued limits $-2\pi \mathrm{i}, -\pi \mathrm{i}, 0$, respectively, for the same specialized function. This is an exact counterexample to an uncompleted independent exponent limit, not an error in the earlier single-resolvent theorem.

At a general negative-total hyperplane $\alpha + \beta = -M$, the coefficient entering the mean residue is explicitly

$$[x^{M-1}] V(x; \alpha, -M - \alpha) = (2\pi \mathrm{i})^{M-1} \binom{\alpha + M - 1}{M - 1}. \tag{2.26}$$

Indeed this coefficient is the residue at zero of $e^{\pi \mathrm{i}(2\alpha+M)x} (\pi / \sin \pi x)^M \, \mathrm{d}x$; putting $w = e^{2\pi \mathrm{i} x}$ converts it to $(2\pi \mathrm{i})^{M-1} \operatorname{Res}_{w=1} w^{\alpha+M-1} (w - 1)^{-M} \, \mathrm{d}w$. Consequently the residue of (2.23) is

$$(1 - e^{2\pi \mathrm{i}\alpha})(2\pi \mathrm{i})^{M-1} \binom{\alpha + M - 1}{M - 1}.$$

For example, at $\alpha = \beta = -1/2$ it is 2, and direct integration of $\pi \csc \pi x$ gives

$$\mathrm{FP}_x \int_0^1 \pi \csc(\pi x)\, \mathrm{d}x = 2 \log(2/\pi). \tag{2.27}$$

The dependence on the path can survive even when every transverse linear path gives the expected zero. At $(-1,-1)$ write

$$\alpha = -1 + (T+D)/2, \qquad \beta = -1 + (T-D)/2.$$

On the curved path $T = cD^2$ with $c \neq 0$, the Gamma quotient (2.23) has limit π^2/c. Indeed its leading terms are $\pi^2 D^2/T - \pi^2 T$. The specialized kernel is $\pi^2 \csc^2 \pi x$, whose coordinate finite part is zero. Its full contact is $(1 - e^{\pi \mathrm{i} D})\pi \mathrm{i} D/T$, with the same curved-path limit. The completed value is again independent of the path.

Corollary 2.7 (A finite-part logarithmic family in ordinary constants)**.** *Let $R_\pm = R_{\pm\pi\mathrm{i}}$ and put*

$$\begin{aligned} Q(a,b) &= e^{-(a+b)\log(2\pi) + \pi\mathrm{i}(a-b)/2} \frac{\Gamma(1+a+b)}{\Gamma(1+a)\Gamma(1+b)}, \\ H(a,b) &= \frac{-2\pi\mathrm{i} a Q(a,b) - 1 + e^{\pi\mathrm{i}(a-b)}}{a+b}. \end{aligned} \tag{2.28}$$

This quotient is holomorphic at $(0,0)$. For integers $r, s \geq 0$,

$$\mathrm{FP}_x \int_0^1 (K - \pi\mathrm{i})(\operatorname{Log} R_+)^r (\operatorname{Log} R_-)^s \, \mathrm{d}x = r! s! [a^r b^s] H(a,b). \tag{2.29}$$

Every such value is a finite polynomial in $\pi\mathrm{i}$, $\log(2\pi)$, and ordinary zeta values. In particular,

$$\begin{aligned} \mathrm{FP}_x \int_0^1 (K - \pi\mathrm{i}) \operatorname{Log} R_+ \, \mathrm{d}x &= \frac{\pi^2}{2} + 2\pi\mathrm{i}\log(2\pi), \\ \mathrm{FP}_x \int_0^1 (K - \pi\mathrm{i})(\operatorname{Log} R_-)^s \, \mathrm{d}x &= \frac{(-\pi\mathrm{i})^{s+1}}{s+1}. \end{aligned} \tag{2.30}$$

Proof. At $\alpha = -1 + a, \beta = b$, (2.23) equals $-2\pi\mathrm{i} a Q(a,b)/(a+b)$. Subtracting the complete paired endpoint term gives (2.28). On $b = -a$, reflection gives $Q(a,-a) = e^{\pi\mathrm{i}a} \sin(\pi a)/(\pi a)$, so the numerator vanishes identically. Analytic division and the endpoint proof above justify all coefficients. The Gamma expansion proves the stated constant class. For an independent check of the second identity, $(\operatorname{Log} R_-)' = K - \pi\mathrm{i}$. The endpoint logarithms are $\log\eta + o(1)$ and $\log\eta - \pi\mathrm{i} + o(1)$; the constant term of their $(s+1)$st-power difference is $(-\pi\mathrm{i})^{s+1}$. □

2.6 All polynomial moments and parameter derivatives

The same beta formula gives an ordinary exponential moment at the special points:

$$\int_0^1 e^{tx} P_{\alpha,\beta}(x) \, \mathrm{d}x = \frac{e^{\pi\mathrm{i}(\alpha-\beta)/2 + t/2} \Gamma(1+\alpha+\beta)}{(2\pi)^{\alpha+\beta} \Gamma(1+\alpha+t/(2\pi\mathrm{i}))\Gamma(1+\beta-t/(2\pi\mathrm{i}))}. \tag{2.31}$$

It is valid for $\Re(\alpha+\beta) > -1$ and every complex t. Away from zeros of its value at $t=0$, its logarithmic derivatives are

$$\begin{aligned} \kappa_1 &= \frac{1}{2} + \frac{\psi(1+\beta) - \psi(1+\alpha)}{2\pi\mathrm{i}}, \\ \kappa_r &= -\frac{\psi^{(r-1)}(1+\alpha) + (-1)^r \psi^{(r-1)}(1+\beta)}{(2\pi\mathrm{i})^r}, \qquad r \geq 2. \end{aligned} \tag{2.32}$$

Thus its nth moment is $M(\alpha,\beta) Y_n(\kappa_1, \ldots, \kappa_n)$, where Y_n is the complete exponential Bell polynomial. At Gamma zeros, differentiate the full reciprocal-Gamma formula (2.31); the factored Bell expression is only its meromorphic rewriting. All mixed exponent derivatives follow by ordinary parameter differentiation in this domain, and by (2.19) at resonances.

At nonnegative integer $\alpha = m, \beta = n$, these formulas give finite generalized-harmonic expressions at every moment order. Explicitly,

$$M(m,n) = \frac{e^{\pi\mathrm{i}(m-n)/2}}{(2\pi)^{m+n}}\binom{m+n}{m}, \qquad \kappa_1 = \frac{1}{2} + \frac{H_n - H_m}{2\pi\mathrm{i}},$$

and for $r \geq 2$,

$$\kappa_r = \frac{(-1)^{r+1}(r-1)!}{(2\pi\mathrm{i})^r}\left\{\zeta(r) - H_m^{(r)} + (-1)^r\left(\zeta(r) - H_n^{(r)}\right)\right\}.$$

Substitution into the Bell polynomial is a finite identity involving only harmonic numbers, zeta values, and the stated prefactor.

Finally, for any fixed-coordinate moment $\mathcal{M}$ in (2.20), differentiation in a resolvent parameter gives

$$\partial_{z_j}\mathcal{M}(\boldsymbol{\lambda}, \mathbf{k}) = \lambda_j \mathcal{M}(\boldsymbol{\lambda} + \mathbf{e}_j, \mathbf{k}) + k_j \mathcal{M}(\boldsymbol{\lambda} + \mathbf{e}_j, \mathbf{k} - \mathbf{e}_j).$$

The usual finite integration-by-parts formula in powers of $\operatorname{Log} R_{z_j}$ therefore supplies a normalized primitive in each z_j. For example, with the other factors fixed and $\lambda_j \neq 1$, a primitive at $k_j = k$ is

$$\sum_{h=0}^{k} \frac{(-1)^h k!}{(k-h)!(\lambda_j - 1)^{h+1}} \mathcal{M}(\boldsymbol{\lambda} - \mathbf{e}_j, \mathbf{k} - h\mathbf{e}_j).$$

When $\lambda_j = 1$ it is the same moment with exponent 0, logarithmic degree $k+1$, divided by $k+1$. Subtract its value at a chosen base point in the same half-plane to fix the additive constant. The proof is coefficientwise differentiation of the completed integrals; no uncompleted singular restriction is used.

3 Mixed logarithmic moments in finite polylogarithmic depth

The general complex-power transform retains an explicit Dirichlet coordinate. Its first mixed exponent derivative admits a substantially smaller expression: two opposite logarithmic resolvents require only depth-two polylogarithms, at every Stieltjes index and every argument derivative. This section proves that reduction with its full finite-part normalization.

3.1 Two and three ordinary logarithmic moments

Take z in the upper half-plane and w in the lower half-plane. Set

$$q = \frac{z - \pi\mathrm{i}}{z + \pi\mathrm{i}}, \qquad r = \frac{w + \pi\mathrm{i}}{w - \pi\mathrm{i}}, \qquad c_z = \operatorname{Log}\left(-\frac{1}{z + \pi\mathrm{i}}\right), \quad c_w = \operatorname{Log}\left(-\frac{1}{w - \pi\mathrm{i}}\right),$$

with the branches fixed in Section 2. Define

$$L_z(x) = \operatorname{Log} R_z(x) - c_z, \qquad L_w(x) = \operatorname{Log} R_w(x) - c_w.$$

Their L^2-convergent Fourier expansions are

$$L_z(x) = \sum_{n\geq 1} \frac{q^n - 1}{n} e^{2\pi\mathrm{i}nx}, \qquad L_w(x) = \sum_{n\geq 1} \frac{r^n - 1}{n} e^{-2\pi\mathrm{i}nx}. \tag{3.1}$$

In particular their means vanish.

Proposition 3.1 (A dilogarithmic covariance)**.** *The ordinary integral is*

$$\int_0^1 L_z(x) L_w(x)\,\mathrm{d}x = \operatorname{Li}_2(qr) - \operatorname{Li}_2(q) - \operatorname{Li}_2(r) + \zeta(2) =: \mathcal{D}(q,r). \tag{3.2}$$

Equivalently, the uncentered integral is $c_z c_w + \mathcal{D}(q,r)$. For two parameters in the same half-plane, the centered integral is zero.

Proof. Pair the two L^2 Fourier series. Their constant product coefficient is $\sum_{n\ge1}(q^n-1)(r^n-1)/n^2$, which is absolutely convergent and sums to (3.2). Two equally oriented series have no zero product mode. The logarithmic endpoint bounds also verify ordinary integrability directly. □

We use the decreasing-index convention

$$\mathrm{Li}_{s_1,\dots,s_d}(u_1,\dots,u_d)=\sum_{n_1>\cdots>n_d\ge1}\frac{u_1^{n_1}\cdots u_d^{n_d}}{n_1^{s_1}\cdots n_d^{s_d}}. \tag{3.3}$$

Individual colors may have modulus greater than one; the actual coupled summands, or equivalently their cumulative products, determine convergence. Continuous limiting values are used when a displayed quotient of colors is $0/0$.

For two upper-half-plane parameters with disk coordinates q_1,q_2 and one lower-half-plane parameter with coordinate r, let $\nu_q(q)=1$, $\nu_q(1)=-1$. Then

$$\begin{aligned}\int_0^1 L_{z_1}(x)L_{z_2}(x)L_w(x)\,\mathrm{d}x&\\ =\sum_{a\in\{q_1,1\}}\sum_{b\in\{q_2,1\}}\sum_{c\in\{r,1\}}&\nu_{q_1}(a)\nu_{q_2}(b)\nu_r(c)\\ &\times\left\{\mathrm{Li}_{2,1}(ac,b/a)+\mathrm{Li}_{2,1}(bc,a/b)\right\}.\end{aligned} \tag{3.4}$$

This follows from the absolutely convergent elementary identity

$$\sum_{n,m\ge1}\frac{x^ny^m}{nm(n+m)}=\mathrm{Li}_{2,1}(x,y/x)+\mathrm{Li}_{2,1}(y,x/y),$$

using $1/(nm(n+m))=1/((n+m)^2n)+1/((n+m)^2m)$. The two cumulative colors are ac,bc, both of modulus at most one. At three zero disk coordinates the value is $-2\zeta(3)$. Formula (3.4) is thus a finite weight-three identity with the branch and color conventions explicit.

3.2 A depth-two transform with an arbitrary spectral order

For $\Re u>0$, define

$$\mathcal{T}_u(x,y)=\mathrm{Li}_{u+1}(x)\,\mathrm{Li}_1(xy)-\mathrm{Li}_{1,u+1}(xy,1/y). \tag{3.5}$$

For $x=y=1$ replace the right side, whose separate terms diverge, by the explicitly convergent value

$$\mathcal{T}_u(1,1)=\zeta(u+1,1)+\zeta(u+2). \tag{3.6}$$

Here $\zeta(s,t)=\sum_{n>m\ge1}n^{-s}m^{-t}$ is strict. At zero colors the limit is used; in particular $\mathcal{T}_u(0,y)=\mathcal{T}_u(x,0)=0$. Set

$$\mathcal{P}_u(q,r)=\mathcal{T}_u(q,r)-\mathcal{T}_u(q,1)-\mathcal{T}_u(1,r)+\mathcal{T}_u(1,1). \tag{3.7}$$

Lemma 3.2 (The two Fourier Dirichlet series)**.** *If $f(x)=L_z(x)L_w(x)$, then*

$$\sum_{k\ge1}\frac{\widehat{f}(k)}{k^u}=\mathcal{P}_u(q,r),\qquad\sum_{k\ge1}\frac{\widehat{f}(-k)}{k^u}=\mathcal{P}_u(r,q),\qquad\Re u>0. \tag{3.8}$$

All sums in this assertion converge absolutely.

Proof. The positive coefficient of the product is

$$\widehat{f}(k)=\sum_{m\ge1}\frac{(q^{m+k}-1)(r^m-1)}{m(m+k)},\qquad k\ge1.$$

The four expanded sums are bounded by a constant multiple of $\sum_{k\ge1} H_k/k^{\Re u+1}$, which converges for $\Re u > 0$. For $|x|, |y| \le 1$ in the required cases,

$$\begin{aligned}\sum_{k,m\ge1} \frac{x^{k+m}y^m}{k^u m(k+m)} &= \sum_{k,m\ge1} \frac{x^k(xy)^m}{k^{u+1}m} - \sum_{n>k\ge1} \frac{(xy)^n y^{-k}}{nk^{u+1}} \\ &= \mathcal{T}_u(x,y).\end{aligned}$$

The first equality uses $1/(m(m+k)) = k^{-1}(m^{-1} - (m+k)^{-1})$. At $(x,y) = (1,1)$ the unseparated sum is $\sum H_k/k^{u+1} = \zeta(u+1,1) + \zeta(u+2)$. Expansion of the four numerator terms gives the first formula in (3.8); exchanging q, r gives the second. □

The expressions in (3.5) are finite combinations of depth at most two. When u is not integral, (3.3) is an explicit multiple Dirichlet sum with a complex order; spectral differentiation inserts logarithms of its integer indices. Meromorphic continuation is always taken for the complete combination, not for a divergent product at $(x,y) = (1,1)$.

3.3 All Stieltjes indices and all argument derivatives

Write the local analytic logarithms

$$\ell_z(x) = -\operatorname{Log}(\pi x \cot \pi x - zx), \qquad \ell_w(x) = -\operatorname{Log}(\pi x \cot \pi x - wx),$$

so that $L_z(x) = \log x - c_z + \ell_z(x)$ and similarly for w. Define the finite coefficients $t_{p,h}$ by

$$[x^p]\{(X - c_z + \ell_z(x))(X - c_w + \ell_w(x))\} = \sum_{h=0}^{2} t_{p,h} X^h. \tag{3.9}$$

For example,

$$\begin{aligned}(t_{0,2}, t_{0,1}, t_{0,0}) &= (1, -c_z - c_w, c_z c_w), \\ (t_{1,2}, t_{1,1}, t_{1,0}) &= (0, z + w, -c_z w - c_w z).\end{aligned}$$

For $p \ge 1$, $t_{p,2} = 0$. Taylor expansion of the cotangent computes every required coefficient exactly.

Theorem 3.3 (Finite depth-two Stieltjes identity)**.** *For $p \ge 0$ the expression*

$$\begin{aligned}\mathcal{E}_p(s;q,r) = \Gamma(s)\left\{\sigma_+^{p-s}\mathcal{P}_{s-p}(q,r) + \sigma_-^{p-s}\mathcal{P}_{s-p}(r,q)\right\} + \mathbf{1}_{p=0}\frac{\mathcal{D}(q,r)}{s} \\ - A_p(s)\sum_{h=0}^{2}\frac{(-1)^h h!\, t_{p,h}}{s^{h+1}}\end{aligned} \tag{3.10}$$

is holomorphic at $s = 0$. For every $m \ge 0$,

$$\boxed{\operatorname{FP}_x \int_0^1 \gamma_m^{(p)}(x) L_z(x) L_w(x)\,\mathrm{d}x = m![s^m]\mathcal{E}_p(s;q,r).} \tag{3.11}$$

In particular the order-m answer uses only finitely many order derivatives of the depth-two functions in (3.5)–(3.7), ordinary zeta functions, Gamma functions, logarithms, and the explicitly finite local coefficients (3.9).

Proof. For $\Re s > p+1$, pairing with the uniformly convergent Hurwitz Fourier series and applying Lemma 3.2 gives the first line of (3.10), before its last subtraction, as the integral of $\partial_x^p(\zeta(1-s,x) + 1/s)L_z L_w$. Its mean contribution is exactly $\mathbf{1}_{p=0}\mathcal{D}/s$ by Proposition 3.1.

At zero, use (2.15). The smooth part times $L_z L_w$ is locally integrable. Among the terms multiplying $A_p(s)x^{s-p-1}$, only the coefficient x^p has a resonant integration exponent. By (3.9), its integrated amplitude is

$$A_p(s)\sum_{h=0}^{2} t_{p,h}\int_0^1 x^{s-1}\log^h x\,\mathrm{d}x = A_p(s)\sum_{h=0}^{2}\frac{(-1)^h h! t_{p,h}}{s^{h+1}}.$$

At $x=1$ the Hurwitz factor is smooth and the logarithmic kernel is integrable, so there is no second resonant term. The finite-subtraction proof of Theorem 2.5 now applies verbatim to these power-log monomials: removing their full amplitudes gives a holomorphic germ whose coefficients are the fixed-coordinate finite parts. This proves both assertions, including analytic continuation of the complete depth-two expression wherever it is needed near $s=0$. □

For $p=0$ the subtraction is particularly transparent:

$$\begin{aligned}\mathcal{E}_0(s;q,r) = \Gamma(s)\left\{(-2\pi\mathrm{i})^{-s}\mathcal{P}_s(q,r) + (2\pi\mathrm{i})^{-s}\mathcal{P}_s(r,q)\right\} + \frac{\mathcal{D}(q,r)}{s}\\ -\frac{2}{s^3} - \frac{c_z+c_w}{s^2} - \frac{c_z c_w}{s}.\end{aligned} \tag{3.12}$$

Keeping only the leading cubic pole would lose two prescribed endpoint constants. For $p>0$, keeping only pole residues would additionally lose the regular coefficients contributed by the polynomial $A_p(s)$.

3.4 A special-point identity involving a harmonic Laurent coefficient

Let $L=\log(2\pi)$ and define η by the strict expansion

$$\zeta(1+s,1) = \frac{1}{s^2} + \frac{\gamma}{s} + \frac{\gamma^2-\zeta(2)}{2} + \eta s + O(s^2). \tag{3.13}$$

This is the same coefficient used in the cubic and quartic incoming reports [4, 3]. At $q=r=0$, $\mathcal{P}_s(0,0) = \zeta(1+s,1)+\zeta(2+s)$ and $\mathcal{D}(0,0)=\zeta(2)$. The kernel is real:

$$L_{\pi\mathrm{i}}(x)L_{-\pi\mathrm{i}}(x) = \log^2(2\sin\pi x) + \pi^2(x-\tfrac12)^2.$$

Expanding (3.12) gives the exact identity

$$\begin{aligned}&\mathrm{FP}_x\int_0^1 \gamma_0(x)\left\{\log^2(2\sin\pi x) + \pi^2(x-\tfrac12)^2\right\}\mathrm{d}x\\ &\quad = 2\eta + 2\zeta'(2) - \frac{\gamma^3+L^3}{3} - \gamma\zeta(2) - \frac{L}{2}\zeta(2) - \frac{2}{3}\zeta(3).\end{aligned} \tag{3.14}$$

For completeness, put $A=\gamma+L$. The first four coefficients of $\Gamma(s)(2\pi)^{-s}\cos(\pi s/2)$ are

$$\frac{1}{s} - A + \left(\frac{A^2}{2} - \frac{\zeta(2)}{4}\right)s + \left(-\frac{A^3}{6} + \frac{A\zeta(2)}{4} - \frac{\zeta(3)}{3}\right)s^2.$$

Multiplication by (3.13) plus $\zeta(2+s)$ proves (3.14). This calculation also provides a direct connection with the harmonic-jet power identities developed below. It does not claim an ordinary-constant evaluation of η.

3.5 A generating identity for every mixed logarithmic moment

At the special resolvent points, the Gamma mean (2.23) gives a compact all-order ordinary identity. For $F_+ = L_{\pi\mathrm{i}}$ and $F_- = L_{-\pi\mathrm{i}}$,

$$\int_0^1 e^{aF_+(x)+bF_-(x)}\,\mathrm{d}x = \frac{\Gamma(1+a+b)}{\Gamma(1+a)\Gamma(1+b)}. \tag{3.15}$$

Initially $\Re(a+b) > -1$; in particular it is analytic at $(0,0)$. Taking its logarithm gives

$$\sum_{n\geq 2} \frac{(-1)^n\zeta(n)}{n}\left((a+b)^n - a^n - b^n\right). \tag{3.16}$$

This power series is used locally, for example when $|a|, |b|, |a+b| < 1$. Consequently every mixed cumulant with $p, q \geq 1$ is

$$(-1)^{p+q}(p+q-1)!\zeta(p+q), \tag{3.17}$$

and the pure centered cumulants vanish. Exponentiating (3.16) is a finite coefficient algorithm for every moment. For example,

$$\begin{aligned} \int F_+F_- &= \zeta(2), & \int F_+^2F_- &= -2\zeta(3),\\ \int F_+^2F_-^2 &= 11\zeta(4), & \int F_+^3F_-^2 &= -24\zeta(5) - 12\zeta(2)\zeta(3). \end{aligned} \tag{3.18}$$

All integrals in (3.18) are ordinary integrals over $(0,1)$. This is a beta/Gamma specialization, with the classical Gamma expansion retaining its attribution [9]; its role here is to evaluate the two-orientation logarithmic moment family exactly.

3.6 Derivatives and primitives in two resolvent parameters

Since $\partial_z c_z = C_z$, differentiation gives $\partial_z L_z = R_z - C_z$. Thus (3.11) also supplies a two-parameter primitive for moments of centered resolvents. For the uncentered version put

$$S_{m,p}(z) = \mathrm{FP}_x \int_0^1 \gamma_m^{(p)}(x)\,\mathrm{Log}\,R_z(x)\,\mathrm{d}x,$$

whose single-logarithm formula is already proved in [2, complex-power section]. Because $\mathrm{FP}\int_0^1 \gamma_m^{(p)} = 0$, the uncentered mixed-logarithm moment is

$$\mathcal{U}_{m,p}(z,w) = m![s^m]\mathcal{E}_p(s;q,r) + c_zS_{m,p}(w) + c_wS_{m,p}(z). \tag{3.19}$$

It satisfies the exact differential identity

$$\partial_z\partial_w\mathcal{U}_{m,p}(z,w) = \mathrm{FP}_x \int_0^1 \gamma_m^{(p)}(x)R_z(x)R_w(x)\,\mathrm{d}x. \tag{3.20}$$

Integrating over a rectangle in the product of the two half-planes therefore yields the alternating four evaluations of $\mathcal{U}_{m,p}$ at its corners, which fixes all integration constants. Partial fractions give the additional consistency check $R_zR_w = (R_z - R_w)/(z-w)$. The differential identity is valid for the completed coordinate finite parts and does not presume that separate uncompleted spectral limits can be multiplied.

4 A finite harmonic reduction for every digamma power

The newest source reports contain two independent fourth-power bridges. The report *Quartic Tails and Complex Resolvents*, Question 1 in its research section, asks for the fifth power with every finite correction[2]; Question 2 asks about polynomials that remove lower-depth data. *Barnes Resolvents and Higher Harmonic Identities*, Question F3, asks for all higher powers and explicitly allows additional harmonic numerators[3]. We answer the fifth-power problem and give a terminating reduction for every fixed power. The latter has a precisely specified output: ordinary convergent multiple zeta values and first Laurent coefficients of strict multiple zeta functions. It does not assert that those coefficients reduce to ordinary Stieltjes constants.

Write

$$H_N(s_1,\dots,s_d)=\sum_{N\ge n_1>\cdots>n_d\ge 1} n_1^{-s_1}\cdots n_d^{-s_d},\qquad Z_d(s_1,\dots,s_d)=\lim_{N\to\infty} H_N(s_1,\dots,s_d)$$

when the latter converges. The larger index is first. The empty word has value one, and $H_N^{(r)}=H_N(r)$, $H_N=H_N^{(1)}$. The classical harmonic products, Euler-sum reductions and digamma expansions keep their existing attribution[11, 12, 13]. All indices in a harmonic word below are positive integers, except for the one explicitly moving outer order. We use

$$\zeta(1+u)=u^{-1}+\sum_{j\ge 0}\frac{(-1)^j\gamma_j}{j!}u^j,\qquad \gamma=\gamma_0,\qquad Q_k=\operatorname{FP}\int_0^1\psi(x)^k\,\mathrm{d}x.$$

Here FP is the constant in the cutoff coordinate $x=\varepsilon$: it is the coefficient of $\varepsilon^0(\log\varepsilon)^0$ in $\int_\varepsilon^1$. The upper endpoint is regular. Consequently

$$\operatorname{FP}\int_0^1 x^{-j}\,\mathrm{d}x=-\frac{1}{j-1}\quad(j\ge 2),\qquad \operatorname{FP}\int_0^1 x^{-1}\,\mathrm{d}x=0.$$

4.1 The coefficients and the uniform theorem

For $n\ge 0$ introduce the formal series

$$V_n(t)=-1+(H_n-\gamma)t+\sum_{r\ge 2}(H_n^{(r)}+(-1)^r\zeta(r))t^r. \tag{4.1}$$

Only finitely many terms are needed in every operation that follows. For $1\le j\le k$ put

$$p_{k,j}(n)=[t^{k-j}]V_n(t)^k,\qquad \kappa_k=[t^k]V_0(t)^k,\qquad \Delta p_{k,j}(n)=p_{k,j}(n)-p_{k,j}(n-1). \tag{4.2}$$

Define

$$B_k=\kappa_k+\sum_{j=2}^{k}\left\{\frac{1}{j-1}\sum_{n\ge 1}\frac{\Delta p_{k,j}(n)}{n^{j-1}}-\sum_{n\ge 1}\frac{p_{k,j}(n)}{n^j}\right\}, \tag{4.3}$$

$$C_k=[L^0]\sum_{n=1}^{N}\frac{p_{k,1}(n)}{n},\qquad L=\log N, \tag{4.4}$$

$$E_k(u)=\sum_{n\ge 1}\Delta p_{k,1}(n)n^{-u}\qquad(\Re u>0). \tag{4.5}$$

The notation in (4.4) means the constant in the polynomial asymptotic expansion in L, with an $o(1)$ remainder; existence is part of the theorem. If $k=1$, the sum in (4.3) is empty.

Theorem 4.1 (Uniform finite harmonic reduction)**.** *For every integer $k\ge 1$, all series in (4.3) converge absolutely, C_k exists, E_k is meromorphic near zero, and*

$$\boxed{Q_k=B_k-C_k+[u^1]E_k(u).} \tag{4.6}$$

Every term on the right has a finite algebraic construction from (4.1). More precisely:

1. *B_k and C_k are finite linear combinations of convergent multiple zeta values, with coefficients in $\mathbb{Q}[\gamma, \zeta(2), \dots, \zeta(k)]$. No limiting harmonic sum remains after the finite reduction given below.*

2. *$E_k(u)$ is a finite linear combination of functions $Z_{d+1}(m+u, s_1, \dots, s_d)$, where all displayed fixed indices are positive integers, $m + s_1 + \cdots + s_d \le k-1$, and $d+1 \le k-1$. Coefficients are in the same ordinary-constant algebra.*

3. *For $k \ge 2$, its only term of depth $k-1$ is*

$$-k!\, Z_{k-1}(1+u, \underbrace{1, \dots, 1}_{k-2}). \tag{4.7}$$

   All other moving-order terms have depth at most $k-2$.

For $k = 1$, $E_1 = 0$ and $Q_1 = 0$. For $k = 2$, the formula gives $Q_2 = 2\gamma_1 - 2\zeta(2)$.

The depth statement describes the specified finite normal form. It is not an arithmetic independence statement. In particular, shuffle, stuffle, duality, or an additional analytic identity may further reduce the retained coordinates.

4.2 The entire remainder and the normalized primitive

Lemma 4.2 (Mittag–Leffler identity at every power)**.** *Let $P_{k,n}(z) = \sum_{j=1}^{k} p_{k,j}(n)(n+z)^{-j}$. Away from the nonpositive integers,*

$$\psi(z)^k = P_{k,0}(z) + \kappa_k + \sum_{n \ge 1} (P_{k,n}(z) - P_{k,n}(0)). \tag{4.8}$$

The series converges normally on compact subsets of that domain.

Proof. The digamma recurrence and its Taylor expansion at zero give $t\psi(-n+t) = V_n(t)$. Therefore (4.2) gives exactly the principal parts, and κ_k is the constant at the origin. To determine the entire remainder, set $f(z) = \psi(z)^k - P_{k,0}(z)$ and integrate $zf(w)/(w(w-z))$ on $|w| = N + 1/2$. Uniformly on those circles, $f(w) = O(\log^k N)$. In the right half-plane this is the usual digamma asymptotic. In the left half-plane use $\psi(w) = \psi(1-w) - \pi\cot(\pi w)$. The cotangent is uniformly bounded on the chosen circles: where $|\Im w| \ge 1/4$ its exponential formula gives the bound; on the remaining arcs $\Re w$ stays a fixed distance from the integers for sufficiently large N. The contour integral is consequently $O(\log^k N/N)$ and tends to zero. Its residues give (4.8), including the constant $f(0) = \kappa_k$.

Finally $p_{k,j}(n) = O(1 + \log^{k-j} n)$ and $(n+z)^{-j} - n^{-j} = O(n^{-j-1})$ on a fixed compact set. This proves normal convergence. Matching principal parts alone would not have proved the lemma: the vanishing contour integral is what fixes the entire function. □

For $n \ge 1$ set

$$F_1(n,z) = \operatorname{Log}(1 + z/n) - z/n, \qquad F_j(n,z) = \frac{(n+z)^{1-j} - n^{1-j}}{1-j} - \frac{z}{n^j} \quad (j \ge 2).$$

On $\Re z > 0$ use the logarithm real on the positive real axis. The lemma immediately gives the following all-power antiderivative.

Corollary 4.3 (A primitive with fixed additive constant)**.** *For every $k \ge 1$,*

$$\mathcal{P}_k(z) = p_{k,1}(0)\operatorname{Log} z + \sum_{j=2}^{k} \frac{p_{k,j}(0) z^{1-j}}{1-j} + \kappa_k z + \sum_{n \ge 1} \sum_{j=1}^{k} p_{k,j}(n) F_j(n,z) \tag{4.9}$$

satisfies $\mathcal{P}_k'(z) = \psi(z)^k$ on $\Re z > 0$. Its expansion at the singular origin has constant zero. Thus $\mathcal{P}_k(b) = \operatorname{FP} \int_0^b \psi(x)^k \, \mathrm{d}x$ for $b > 0$, and ordinary integrals between positive endpoints are differences of $\mathcal{P}_k$. Every fixed argument derivative of (4.8) is also justified by normal convergence.

Proof. The displayed F_j are the primitives, vanishing at zero, of $(n+z)^{-j}-n^{-j}$. Their size on a fixed compact set is $O(n^{-j-1})$; the same normal-convergence argument applies to the differentiated series. On $|z|<1$ all nonsingular summands vanish at zero. This proves the normalization as well as the derivative identity. □

4.3 Proof of the finite reduction

Integrate (4.8). For each $j\geq 2$ finite summation by parts and the vanishing terminal boundary term give

$$-\frac{p_{k,j}(0)}{j-1}+\sum_{n\geq 1}p_{k,j}(n)\left\{\frac{n^{1-j}-(n+1)^{1-j}}{j-1}-n^{-j}\right\}$$
$$=\frac{1}{j-1}\sum_{n\geq 1}\frac{\Delta p_{k,j}(n)}{n^{j-1}}-\sum_{n\geq 1}\frac{p_{k,j}(n)}{n^{j}}.$$

All these sums are absolutely convergent. Hence

$$Q_k=B_k+\sum_{n\geq 1}p_{k,1}(n)\left\{\log(1+1/n)-1/n\right\}. \tag{4.10}$$

Write $p(n)=p_{k,1}(n)$ and $d(n)=p(n)-p(n-1)$. At a finite cutoff,

$$\sum_{n=1}^{N}p(n)\left\{\log(1+1/n)-1/n\right\}=p(N)\log(N+1)-\sum_{n=1}^{N}d(n)\log n-\sum_{n=1}^{N}\frac{p(n)}{n}. \tag{4.11}$$

No divergent infinite rearrangement has been used.

The harmonic asymptotics imply, for a polynomial P, $p(N)=P(\log N)+O(N^{-1}\log^{M}N)$. Thus the first term on the right of (4.11) has zero constant in $L=\log N$. Likewise

$$d(n)=\frac{\sum_{r=0}^{k-2}a_r\log^r n}{n}+O(n^{-2}\log^{M}n) \tag{4.12}$$

for some coefficients a_r (the case of a zero polynomial is allowed). Consequently

$$E_k(u)=\sum_{r}a_r(-1)^r\zeta^{(r)}(1+u)+\sum_{n\geq 1}\left(d(n)-\frac{\sum_r a_r\log^r n}{n}\right)n^{-u},$$

where the second sum is holomorphic for $\Re u>-1$, locally normally with every fixed derivative. The Stieltjes coefficient convention gives

$$[u^1]E_k(u)=-[L^0]\sum_{n=1}^{N}d(n)\log n. \tag{4.13}$$

Indeed $[u^1](-1)^r\zeta^{(r)}(1+u)=-\gamma_{r+1}$, while the remaining coefficient is minus its absolutely convergent logarithmic sum. Since (4.10) converges, taking constants in (4.11) proves (4.6).

Here are finite rules that compute every object in that proof. Expand each harmonic polynomial by the strict quasi-shuffle rule

$$H_N(a)H_N(b,\mathbf{v})=\sum_{\mathbf{w}\in(a)*(b,\mathbf{v})}H_N(\mathbf{w}), \tag{4.14}$$

where the recursive product inserts either initial letter or merges them to their sum; repeated words retain their multiplicities. The merge has a plus sign. For any nonempty word $\mathbf{w}=(w_1,\dots,w_d)$,

$$\sum_{n=1}^{N}\frac{H_{n-1}(\mathbf{w})}{n^{j}}=H_N(j,\mathbf{w}), \tag{4.15}$$

$$\sum_{n=1}^{N}\frac{H_{n}(\mathbf{w})}{n^{j}}=H_N(j,\mathbf{w})+H_N(j+w_1,w_2,\dots,w_d). \tag{4.16}$$

For the empty word the second term in (4.16) is absent. The differences $\Delta p_{k,j}$ are computed by replacing $H_n^{(r)}$ by $H_{n-1}^{(r)} + n^{-r}$ and subtracting. Equations (4.15)–(4.16) then reduce B_k to convergent multiple zeta values, because its initial index is always at least two. They reduce E_k to the finite moving-order combination in the theorem.

For completeness, leading ones in C_k can also be removed by an explicit terminating rule. Let $\mathcal{R}_{\mathbf{v}}(T)$ denote the harmonic regularization polynomial, with

$$\mathcal{R}_{\varnothing}(T) = 1, \quad \mathcal{R}_{(1)}(T) = T, \quad \mathcal{R}_{\mathbf{v}}(T) = Z(\mathbf{v}) \quad (v_1 \geq 2).$$

For $\mathbf{v} = (1, \mathbf{w})$ with a initial ones, write $(1) * \mathbf{w} = a\mathbf{v} + \sum_{\mathbf{u} \neq \mathbf{v}} m_{\mathbf{u}} \mathbf{u}$. Then

$$\mathcal{R}_{\mathbf{v}}(T) = \frac{1}{a} \left\{ T\mathcal{R}_{\mathbf{w}}(T) - \sum_{\mathbf{u} \neq \mathbf{v}} m_{\mathbf{u}} \mathcal{R}_{\mathbf{u}}(T) \right\}. \tag{4.17}$$

This decreases either total weight or the number of initial ones, so it terminates. Before passing to $\mathcal{R}$, the same recurrence with each terminal $Z(\mathbf{v})$ replaced by $H_N(\mathbf{v})$ is an exact finite identity, with $T = H_N$. Only afterward are the convergent terminal words replaced by their limits. Their tails are $O(N^{-1} \log^M N)$, and $H_N = L + \gamma + O(N^{-1})$. Thus their constant in L is obtained by $T = \gamma$. This proves existence and the finite construction of C_k. It also proves the individual continuation of every moving-order term: when $m = 1$, its inner harmonic word is a polynomial in $\log n$ plus $O(n^{-1} \log^M n)$, so a finite sum of zeta derivatives subtracts its pole. When $m \geq 2$, its defining series is already holomorphic near $u = 0$.

The index-weight bound follows directly from (4.1). Finally, the unique harmonic monomial of degree $k-1$ in the simple-pole coefficient is $-kH_n^{k-1}$. Its difference contributes $-k(k-1)H_{n-1}^{k-2}/n$. The coefficient of the strict word of $k-2$ ones in H_{n-1}^{k-2} is $(k-2)!$. Every other term has smaller harmonic degree after taking a difference. This proves (4.7) and completes the theorem.

5 The fifth-power identity and its additional spectral coordinate

Define the inherited strict coefficients and the new height-one coefficient by

$$\eta = [u^1] Z_2(1+u, 1), \qquad \theta = [u^1] Z_3(1+u, 1, 1), \qquad \phi = [u^1] Z_4(1+u, 1, 1, 1). \tag{5.1}$$

Brackets denote Laurent coefficients, with neither an additional factorial nor a sign. A convergent depth-two derivative coordinate is

$$\begin{aligned} \chi_4 &= 2(\partial_1 - \partial_2) Z_2(3, 1) + \partial_1 Z_2(2, 2) \\ &= -\sum_{n > m \geq 1} \left\{ \frac{2 \log(n/m)}{n^3 m} + \frac{\log n}{n^2 m^2} \right\}. \end{aligned} \tag{5.2}$$

Both sums converge absolutely. The symbols ϕ and χ_4 specify the exact data retained below. There is no assertion that either is arithmetically independent of the other constants.

Theorem 5.1 (Fifth harmonic bridge)**.** *The coordinate finite part of the fifth digamma power is*

$$\boxed{\begin{aligned} Q_5 = {} & 30\gamma^2\zeta(2) - 60\gamma\zeta(3) + \frac{35}{3}\zeta(4) - 120\phi + 120\gamma\theta + 60(\zeta(2) - \gamma^2)\eta \\ & - 20(\gamma^3 - 3\gamma\zeta(2) + 2\zeta(3))\gamma_1 + 10\chi_4 - 20\zeta'(4) + 10\zeta(2)\zeta'(2) - 20\gamma\zeta'(3). \end{aligned}} \tag{5.3}$$

An ordinary absolutely convergent series for the same constant is

$$\boxed{\begin{aligned} Q_5 = {} & 30\gamma^2\zeta(2) - 60\gamma\zeta(3) + \frac{35}{3}\zeta(4) \\ & + 20\gamma_4 - 120\zeta(2)\gamma_2 - 5\sum_{n \geq 1} \frac{\mathcal{U}(n) \log n}{n}. \end{aligned}} \tag{5.4}$$

where, with $p=\psi(n)$ *and* $\ell=\log n$,

$$\mathcal{U}(n)=-4p^3+24\zeta(2)p-12p\psi'(n)-2\psi''(n)+4\ell^3-24\zeta(2)\ell+\frac{4\zeta(2)-2\psi'(n)}{n}+\frac{4p}{n^2}. \tag{5.5}$$

The summand in (5.4) *is* $O(n^{-2}\log^3 n)$.

5.1 Explicit principal parts and every finite constant

Put

$$b_n=H_n-\gamma,\quad c_n=H_n^{(2)}+\zeta(2),\quad d_n=H_n^{(3)}-\zeta(3),\quad e_n=H_n^{(4)}+\zeta(4).$$

The principal coefficients supplied by (4.2) are

$$\begin{aligned} p_{5,5}&=-1, \qquad p_{5,4}=5b_n, \qquad p_{5,3}=5c_n-10b_n^2,\\ p_{5,2}&=10b_n^3-20b_nc_n+5d_n, \qquad p_{5,1}=5r_n,\\ r_n&=-b_n^4+6b_n^2c_n-2c_n^2-4b_nd_n+e_n. \end{aligned} \tag{5.6}$$

The constant at zero is

$$\kappa_5=-\gamma^5+20\gamma^3\zeta(2)-30\gamma^2\zeta(3)-30\gamma\zeta(2)^2+20\gamma\zeta(4)+20\zeta(2)\zeta(3)-5\zeta(5). \tag{5.7}$$

The exact difference of the residue is particularly simple:

$$r_n-r_{n-1}=\frac{4a_{n-1}}{n}+\frac{2c_{n-1}}{n^2}+\frac{4b_{n-1}}{n^3}, \qquad a_n=-b_n^3+3b_nc_n-d_n. \tag{5.8}$$

It follows by substituting $b\mapsto b+n^{-1}$, $c\mapsto c+n^{-2}$, $d\mapsto d+n^{-3}$, and $e\mapsto e+n^{-4}$ into r. In particular, it is an exact finite identity before any analytic continuation.

The lower-pole constant in (4.3) is

$$\begin{aligned} B_5=&-\gamma^5+30\gamma^3\zeta(2)+30\gamma^2\zeta(2)-80\gamma^2\zeta(3)-60\gamma\zeta(3)-\frac{65}{2}\gamma\zeta(4)\\ &+60\zeta(2)\zeta(3)+\frac{35}{3}\zeta(4)-69\zeta(5). \end{aligned} \tag{5.9}$$

We provide finite bookkeeping for this reduction and for the second constant, so that the result is not dependent on a numerical fit. The convergent values of weights at most five needed after the quasi-shuffle expansion are

$$\begin{aligned} Z_2(2,1)&=\zeta(3), & Z_2(3,1)&=\tfrac14\zeta(4), & Z_2(2,2)&=\tfrac34\zeta(4),\\ Z_3(2,1,1)&=\zeta(4), & Z_4(2,1,1,1)&=\zeta(5),\\ Z_2(4,1)&=Z_3(3,1,1)=2\zeta(5)-\zeta(2)\zeta(3),\\ Z_2(3,2)&=Z_3(2,2,1)=3\zeta(2)\zeta(3)-\tfrac{11}{2}\zeta(5),\\ Z_2(2,3)&=Z_3(2,1,2)=\tfrac92\zeta(5)-2\zeta(2)\zeta(3). \end{aligned} \tag{5.10}$$

These are classical Euler-sum and duality identities. For example, Euler's formula for $Z_2(n,1)$ gives the $(4,1)$ value; the stuffle product for $\zeta(2)\zeta(3)$ and its iterated-integral shuffle product

$$\zeta(2)\zeta(3)=Z_2(2,3)+3Z_2(3,2)+6Z_2(4,1)$$

give the two remaining depth-two values. Reversing the iterated integral word gives the three indicated dualities. The height-one values follow from the ordinary logarithmic beta integral. Also $\zeta(2)^2=5\zeta(4)/2$.

For an additional check on (5.9), let $B_{5,j}$ denote its summand indexed by j in (4.3), excluding κ_5. The finite rules above give

$$\begin{aligned}
B_{5,2} &= 10\gamma^3\zeta(2) + 30\gamma^2\zeta(2) - 60\gamma^2\zeta(3) - 70\gamma\zeta(3) + \tfrac{85}{2}\gamma\zeta(4) \\
&\qquad + 65\zeta(2)\zeta(3) + 15\zeta(4) - \tfrac{215}{2}\zeta(5), \\
B_{5,3} &= 10\gamma^2\zeta(3) + 10\gamma\zeta(3) - 25\gamma\zeta(4) - 30\zeta(2)\zeta(3) - 5\zeta(4) + \tfrac{115}{2}\zeta(5), \\
B_{5,4} &= 5\gamma\zeta(4) + 5\zeta(2)\zeta(3) + \tfrac{5}{3}\zeta(4) - 15\zeta(5), \\
B_{5,5} &= \zeta(5).
\end{aligned} \tag{5.11}$$

Adding these to (5.7) proves (5.9).

To specify the constant of the simple-pole sum, set

$$A = 4\gamma^3 - 12\gamma\zeta(2) + 4\zeta(3), \quad C_0 = -\gamma^4 + 6\gamma^2\zeta(2) - 4\gamma\zeta(3) - 2\zeta(2)^2 + \zeta(4), \quad e_j(N) = H_N(\underbrace{1,\dots,1}_{j}).$$

Expanding r_n in strict words and applying (4.16) gives the exact finite identity

$$\begin{aligned}
\sum_{n=1}^{N} \frac{r_n}{n} = {} & -24e_5(N) + 24\gamma e_4(N) + 12(\zeta(2) - \gamma^2)e_3(N) + Ae_2(N) + C_0 H_N \\
& - 24H_N(2,1,1,1) + 24\gamma H_N(2,1,1) + 12(\zeta(2) - \gamma^2)H_N(2,1) + AH_N^{(2)} \\
& + 4H_N(1,1,3) + 2H_N(1,2,2) + 4H_N(1,3,1) \\
& + 4H_N(2,3) + 2H_N(3,2) + 4H_N(4,1) \\
& - 4\gamma\{H_N(1,3) + H_N^{(4)}\} + 2\zeta(2)\{H_N(1,2) + H_N^{(3)}\}.
\end{aligned} \tag{5.12}$$

One can apply (4.17) directly. To make its asymptotic evaluation especially short, write $T = H_N$ and note

$$\begin{aligned}
H_N(1,3) &= T\zeta(3) - \tfrac{5}{4}\zeta(4) + o(1), \\
H_N(1,2) &= T\zeta(2) - 2\zeta(3) + o(1), \\
4H_N(1,1,3) + 2H_N(1,2,2) + 4H_N(1,3,1) &= 2T^2\zeta(3) - \tfrac{5}{2}T\zeta(4) - 2\zeta(2)\zeta(3) + o(1).
\end{aligned}$$

Every $o(1)$ remains so after the displayed polynomial logarithmic factors. The last line follows by stuffling H_N into $H_N(1,3)$, $H_N(2,2)$ and $H_N(3,1)$ and then using (5.10). The elementary symmetric polynomials satisfy

$$e_j(N) = [t^j] \exp\left(Tt + \sum_{r=2}^{j} \frac{(-1)^{r-1} H_N^{(r)}}{r} t^r \right).$$

Substitute these formulas and $T = L + \gamma + o(1)$ into (5.12). The complete result is

$$\sum_{n=1}^{N} \frac{r_n}{n} = -\frac{L^5}{5} + 4\zeta(2)L^3 - 18\zeta(4)L + C_r + o(1), \tag{5.13}$$

where

$$C_r = -\frac{\gamma^5}{5} + 6\gamma^3\zeta(2) - 16\gamma^2\zeta(3) - \frac{13}{2}\gamma\zeta(4) + 12\zeta(2)\zeta(3) - \frac{69}{5}\zeta(5). \tag{5.14}$$

Since $p_{5,1} = 5r$, $C_5 = 5C_r$. Thus the substantial ordinary-constant cancellation is exactly

$$B_5 - C_5 = 30\gamma^2\zeta(2) - 60\gamma\zeta(3) + \frac{35}{3}\zeta(4). \tag{5.15}$$

In particular, neither the finite constant at infinity nor the finite parts of the higher poles have been discarded.

5.2 The moving-order identity and the proof of the bridge

Let $E_r(u) = E_5(u)/5$. From (5.8), finite strict harmonic identities give the full meromorphic germ

$$\begin{aligned} E_r(u) = &-24Z_4(1+u,1,1,1) + 24\gamma Z_3(1+u,1,1) + 12(\zeta(2)-\gamma^2)Z_2(1+u,1) \\ &+ 4(\gamma^3 - 3\gamma\zeta(2) + \zeta(3))\zeta(1+u) + 4Z_2(1+u,3) \\ &+ 2Z_2(2+u,2) + 2\zeta(2)\zeta(2+u) + 4Z_2(3+u,1) - 4\gamma\zeta(3+u). \end{aligned} \tag{5.16}$$

For example the identity used for its first line is

$$a_n = -6H_n(1,1,1) + 6\gamma H_n(1,1) + 3(\zeta(2)-\gamma^2)H_n + \gamma^3 - 3\gamma\zeta(2) + \zeta(3) + H_n^{(3)}.$$

The series proof of (5.16) first applies in $\Re u > 0$ and then continues to the germ at zero. It is therefore legitimate to extract its linear coefficient.

The only apparently singular additional depth-two term is removed by an identity valid with a free outer order:

$$Z_2(1+u,3) = \zeta(1+u)\zeta(3) - \zeta(4+u) - Z_2(3,1+u).$$

Consequently

$$[u^1]Z_2(1+u,3) = -\zeta(3)\gamma_1 - \zeta'(4) - \partial_2 Z_2(3,1).$$

All remaining additional terms in (5.16) are ordinary convergent derivatives. Extracting the coefficient and using (5.15) in (4.6) proves (5.3).

For the independently useful convergent form, the elementary polygamma relations give

$$b_{n-1} = \psi(n), \qquad c_{n-1} = 2\zeta(2) - \psi'(n), \qquad d_{n-1} = \tfrac{1}{2}\psi''(n).$$

Substitution into (5.8) shows that

$$r_n - r_{n-1} = \frac{-4\log^3 n + 24\zeta(2)\log n + \mathcal{U}(n)}{n}.$$

The standard positive-axis asymptotics of ψ, ψ', ψ'' give $\mathcal{U}(n) = O(n^{-1}\log^2 n)$. The Stieltjes finite limits are

$$[L^0]\sum_{n=1}^{N}\frac{\log^j n}{n} = \gamma_j.$$

Using them in (4.13) proves (5.4), with ordinary absolute convergence as claimed.

5.3 The highest-depth coefficient and a finite cancellation

The normalization of ϕ has a direct elementary Mellin description. In $\Re s > 1$,

$$Z_4(s,1,1,1) = \frac{1}{6\Gamma(s)}\int_0^\infty \frac{t^{s-1}[-\log(1-e^{-t})]^3}{e^t-1}\,\mathrm{d}t. \tag{5.17}$$

Subtract $(-\log t)^3/(6t)$ on $(0,1)$. The remainder is holomorphic at $s=1$, and its value there is zero, because the unsubtracted kernel is $-1/24$ times the derivative of $[-\log(1-e^{-t})]^4$. Let $g_5 = [u^5]\Gamma(1+u)^{-1}$. Then

$$\begin{aligned} \phi = g_5 &+ \frac{1}{6}\int_0^1 \log t\left\{\frac{[-\log(1-e^{-t})]^3}{e^t-1} - \frac{(-\log t)^3}{t}\right\}\mathrm{d}t \\ &+ \frac{1}{6}\int_1^\infty \frac{\log t\,[-\log(1-e^{-t})]^3}{e^t-1}\,\mathrm{d}t, \end{aligned} \tag{5.18}$$

where both integrals converge ordinarily, and

$$g_5 = \frac{\gamma^5 - 10\gamma^3\zeta(2) + 20\gamma^2\zeta(3) + 15\gamma\zeta(2)^2 - 30\gamma\zeta(4) - 20\zeta(2)\zeta(3) + 24\zeta(5)}{120}.$$

This fixes the depth-four coordinate independently of a formal Laurent label. It does not evaluate it in a smaller constant algebra.

The previously proved cubic and quartic identities[4, 2] are

$$\begin{aligned} Q_3 &= 3\zeta(2) - 6\eta - 6\gamma\gamma_1, \\ Q_4 &= 12\zeta(3) - 12\gamma\zeta(2) + 18\zeta(4) + 4\zeta'(3) - 24\theta + 24\gamma\eta + 12(\gamma^2 - \zeta(2))\gamma_1. \end{aligned}$$

Combining them with the new theorem eliminates both inherited strict coefficients:

Corollary 5.2 (Eliminating the inherited height-one data)**.** *For* $P_5(T) = T^5 + 5\gamma T^4 + 10(\gamma^2 + \zeta(2))T^3$,

$$\begin{aligned} \mathrm{FP}\int_0^1 P_5(\psi(x))\,\mathrm{d}x = &-120\phi + 10\chi_4 + \left(\frac{260}{3} + 90\gamma\right)\zeta(4) - 20\zeta'(4) + 10\zeta(2)\zeta'(2) \\ &- 20(\gamma^3 + 3\gamma\zeta(2) + 2\zeta(3))\gamma_1. \end{aligned} \tag{5.19}$$

The extra derivative coordinate χ_4 remains. Thus this is a proved cancellation of η and θ, not a claimed solution of the stronger all-power, single-coordinate polynomial problem.

5.4 Verification and the exact scope of the result

The accompanying code records sixty exact finite checks. It expands (4.1) exactly, verifies (5.8) and (5.12) on finite harmonic sums, replays the terminating quasi-shuffle regularization, and reproduces (5.9)–(5.15). Its numerical comparison uses two analytically different methods: local integration of the Taylor series of $(x\psi(x))^5$ followed by ordinary quadrature, and (5.4) with a finite asymptotic tail summed through Hurwitz-zeta order derivatives. The direct evaluator gives

$$Q_5 = -8.690303962624538266295589662365653576770293544923884292 66\ldots.$$

The recorded run agrees through forty-eight decimal places. It is a floating-point diagnostic, not an interval certificate; the exact proof is (4.6) together with the finite coefficient calculations above. No numerical relation search is used to infer a constant reduction.

The general theorem settles the existence of a finite reduction for every power in its stated multiple-zeta-jet class and fixes all entire and endpoint constants. The fifth-power theorem supplies the explicit case requested in both newest reports. The arithmetic reduction of χ_4 or ϕ, a proof that a polynomial in ψ can remove all lower-depth data at arbitrary order, and a common-Hurwitz-shift counterpart remain separate research problems.

6 Polylogarithmic counterterms for higher Stieltjes collisions

The geometric collision theorem at Stieltjes index zero in the incoming *Higher Reflection Identities and Resonant Zeta Calculus*[5] leaves two explicit gaps: logarithmic Stieltjes singularities, and weighted collision formulas. The theorem below covers both, together with arbitrary ordinary argument derivatives. The counterterm is finite and explicitly reducible to iterated polylogarithms. The proof also explains why no comparison between different shrinking gaps is needed. It does not evaluate the global merged cubic constant, and it does not assert associativity of a different, successively chosen subtraction prescription.

6.1 Local singular polynomials and the finite-part convention

Normalize generalized Stieltjes functions by

$$\zeta(1+u,a)=\frac{1}{u}+\sum_{m\ge 0}\frac{(-1)^m}{m!}\gamma_m(a)u^m.$$

For $m,r\ge 0$, put

$$f_{m,r}(x)=\gamma_m^{(r)}(\{x\}),\qquad h_{m,r}(z)=\gamma_m^{(r)}(1+z), \tag{6.1}$$

$$d_r=r+1,\qquad L_{m,r}(X)=\prod_{j=1}^{r}(\partial_X-j)X^m. \tag{6.2}$$

The empty product is the identity. Thus $L_{m,0}(X)=X^m$ and $L_{0,r}(X)=(-1)^r r!$. The exact one-sided local decomposition is

$$f_{m,r}(x)=\mathbf{1}_{x>0}\,x^{-r-1}L_{m,r}(\log x)+h_{m,r}(x),\qquad -1<x<1,\quad x\neq 0. \tag{6.3}$$

Derivatives in this formula are ordinary derivatives on the open intervals. They are not derivatives of already extended periodic distributions.

For completeness, the singular polynomial has the harmonic form

$$L_{m,r}(X)=(-1)^r r!\sum_{j=0}^{\min(m,r)}(-1)^j\binom{m}{j}j!e_j\left(1,\frac{1}{2},\dots,\frac{1}{r}\right)X^{m-j}, \tag{6.4}$$

where e_j is an elementary symmetric polynomial and $e_0=1$. Indeed,

$$L_{m,r}(X)=(-1)^r m![z^m](1-z)_r e^{zX}.$$

All analytic tails in (6.3) are holomorphic near zero. Their Taylor coefficients require only ordinary Stieltjes constants and zeta jets: for $k\ge 1$,

$$\gamma_m^{(k)}(1)=(-1)^{m+k}m![u^m](1+u)_k\zeta(k+1+u). \tag{6.5}$$

This follows by differentiating the Hurwitz identity $\partial_a^k\zeta(s,a)=(-1)^k(s)_k\zeta(s+k,a)$.

Fix a periodic smooth weight w which is real analytic in a neighborhood of 0; $w=1$ and $w(x)=e^{2\pi inx}$ are allowed. Fix indices (m_i,r_i), $1\le i\le d$. For distinct sufficiently small real shifts s_i, define

$$C^w_{\boldsymbol{m},\boldsymbol{r}}(\boldsymbol{s})=\mathrm{FP}\int_{\mathbb{R}/\mathbb{Z}}w(x)\prod_{i=1}^{d}f_{m_i,r_i}(x+s_i)\,\mathrm{d}x,\qquad Q^w_{\boldsymbol{m},\boldsymbol{r}}=\mathrm{FP}\int_0^1 w(x)\prod_{i=1}^{d}\gamma_{m_i}^{(r_i)}(x)\,\mathrm{d}x. \tag{6.6}$$

At each singular point a, the finite part removes $(a,a+\eta)$ and retains the constant term in the ambient coordinate η. Distinct points use independent cutoffs. The same convention is used at the merged endpoint. In particular, no cutoff is divided by a shrinking separation.

6.2 A finite Hermite construction

For a nonempty subset $S\subseteq\{1,\dots,d\}$, write

$$d_i=r_i+1,\quad N_S=\sum_{i\in S}d_i,\quad b_S=\min_{i\in S}s_i,$$

$$q_S(x;\boldsymbol{s})=w(x)\prod_{j\notin S}h_{m_j,r_j}(x+s_j),\qquad D_S(x;\boldsymbol{s})=\prod_{i\in S}(x+s_i)^{d_i}, \tag{6.7}$$

$$\Lambda_S(x;\boldsymbol{s})=\prod_{i\in S}L_{m_i,r_i}(\log(x+s_i)),\qquad x>-b_S. \tag{6.8}$$

The polynomial $D_S(x;\boldsymbol{s})$ is monic of degree N_S. Let $P_S(x;\boldsymbol{s})$ be the unique polynomial of degree less than N_S that agrees with q_S at $x=-s_i$ through derivative order d_i-1, for every $i\in S$.

Definition 6.1 (Canonical logarithmic collision counterterm)**.** For $|S| \geq 2$, define

$$T_S(\boldsymbol{s}) = \mathrm{FP}_{R\to\infty}\,\mathrm{FP}_{\eta\downarrow 0}\int_{-b_S+\eta}^{R} \frac{P_S(x;\boldsymbol{s})}{D_S(x;\boldsymbol{s})}\Lambda_S(x;\boldsymbol{s})\,\mathrm{d}x, \qquad T(\boldsymbol{s}) = \sum_{|S|\geq 2} T_S(\boldsymbol{s}). \tag{6.9}$$

The lower finite part is taken first, with the ambient displacement η. At infinity, discard the nonconstant polynomial in $\log R$ and retain its constant term. These are two separate endpoint operations, not a diagonal spectral regularization.

The upper operation in (6.9) is explicit. Put

$$p_S(\boldsymbol{s}) = [x^{N_S-1}]P_S(x;\boldsymbol{s}), \qquad \lambda_S(X) = \prod_{i\in S} L_{m_i,r_i}(X), \qquad \Psi_S(X) = \int_0^X \lambda_S(t)\,\mathrm{d}t.$$

The only divergent upper term is $p_S(\boldsymbol{s})\Psi_S(\log R)$. The difference between its derivative and the integrand is $O(R^{-2}(1+\log R)^{\sum_{i\in S} m_i})$. For a singleton S, the same definition gives zero, since translation to its sole endpoint leaves a sum of pure monomials $x^{-k}(\log x)^j$, $k \geq 1$, whose two-endpoint finite parts vanish.

Theorem 6.2 (All indices, all ordinary derivatives, and analytic weights)**.** *For the convention* (6.6)*, the counterterm* (6.9) *satisfies*

$$C^w_{\boldsymbol{m},\boldsymbol{r}}(\boldsymbol{s}) - T(\boldsymbol{s}) \longrightarrow Q^w_{\boldsymbol{m},\boldsymbol{r}} \qquad (\boldsymbol{s}\longrightarrow 0) \tag{6.10}$$

through distinct real shifts. The convergence imposes no lower bound on ratios of the gaps. In particular, it holds for arbitrary hierarchical collisions and for paths whose relative gaps vanish faster than every power of their overall size.

Every T_S is a finite combination of rational functions of the gaps, values and derivatives of w and the analytic Stieltjes tails at the shifted nodes, and generalized polylogarithms of weight at most $1+\sum_{i\in S} m_i$. All logarithms in its defining real integral have positive arguments. If every $m_i = 0$, the formula is exactly the previous rational–logarithmic collision counterterm, including its ordinary-derivative extension.

Proof of the collision limit. Choose $\delta > 0$ so small that w and all required tails are holomorphic on $|z| < 3\delta$, and restrict $|s_i| < \delta/8$. Use the circle coordinate $(-\delta, 1-\delta)$ and separate the local interval $(-\delta,\delta)$. The outside integral is continuous in the shifts (and analytic when the outside weight is analytic). The local product expands by the subset S of factors contributing their singular terms in (6.3). The empty subset is analytic. The nonempty subset is

$$\mathrm{FP}\int_{-b_S}^{\delta} \frac{q_S(x;\boldsymbol{s})}{D_S(x;\boldsymbol{s})}\Lambda_S(x;\boldsymbol{s})\,\mathrm{d}x. \tag{6.11}$$

Only the endpoint $-b_S$ is singular in this integral: all other nodes $-s_i$ of the subset lie to its left.

Both P_S and the divided remainder

$$A_S(x;\boldsymbol{s}) = \frac{q_S(x;\boldsymbol{s}) - P_S(x;\boldsymbol{s})}{D_S(x;\boldsymbol{s})}$$

extend holomorphically through coalescing nodes. This assertion is not based on individually singular Lagrange coefficients. It follows from the fixed-contour formulas

$$P_S(x;\boldsymbol{s}) = \frac{1}{2\pi i}\oint_{|z|=2\delta} \frac{q_S(z;\boldsymbol{s})}{D_S(z;\boldsymbol{s})}\frac{D_S(z;\boldsymbol{s}) - D_S(x;\boldsymbol{s})}{z-x}\,\mathrm{d}z, \tag{6.12}$$

$$A_S(x;\boldsymbol{s}) = \frac{1}{2\pi i}\oint_{|z|=2\delta} \frac{q_S(z;\boldsymbol{s})}{D_S(z;\boldsymbol{s})(z-x)}\,\mathrm{d}z, \qquad |x| < \delta. \tag{6.13}$$

These also give a uniform bound for A_S on the local integration interval after decreasing δ or the shift neighborhood slightly.

The finite-part tail of the principal part has the convergent form

$$\begin{aligned} B_S(\boldsymbol{s}) &:= \operatorname{FP}\int_\delta^\infty \frac{P_S(x;\boldsymbol{s})}{D_S(x;\boldsymbol{s})}\Lambda_S(x;\boldsymbol{s})\,\mathrm{d}x \\ &= -p_S(\boldsymbol{s})\Psi_S(\log\delta) + \int_\delta^\infty \left\{\frac{P_S(x;\boldsymbol{s})}{D_S(x;\boldsymbol{s})}\Lambda_S(x;\boldsymbol{s}) - \frac{p_S(\boldsymbol{s})}{x}\lambda_S(\log x)\right\}\,\mathrm{d}x. \end{aligned} \tag{6.14}$$

For $x \ge \delta$ use the holomorphic branches of $\operatorname{Log}(x+s_i)$ which are real at real shifts. The integrand and its fixed parameter derivatives are locally uniformly $O(x^{-2}(1+\log x)^M)$, with $M = \sum_{i\in S} m_i$. Consequently B_S is holomorphic near the full diagonal.

Subtracting T_S from (6.11) leaves exactly

$$\int_{-b_S}^{\delta} A_S(x;\boldsymbol{s})\Lambda_S(x;\boldsymbol{s})\,\mathrm{d}x - B_S(\boldsymbol{s}). \tag{6.15}$$

The rational denominators have disappeared from the remaining local integral. To prove the required uniformity, set $y = x + b_S$ and $a_i = s_i - b_S \ge 0$. Its integrand becomes

$$A_S(y - b_S;\boldsymbol{s})\prod_{i\in S} L_{m_i,r_i}(\log(y+a_i)).$$

For δ small, all $y + a_i$ in this integral are less than 1. Because $y + a_i \ge y$, its absolute value is bounded by $K(1+|\log y|)^M$, independently of every gap ratio. This is integrable at $y = 0$. On $y \ge \epsilon > 0$ the convergence is uniform as all shifts vanish; the integral over $0 < y < \epsilon$ is uniformly small as $\epsilon \downarrow 0$. Hence (6.15) tends to

$$\int_0^\delta A_S(x;0)\lambda_S(\log x)\,\mathrm{d}x - B_S(0).$$

Here $P_S(x;0)$ is the Taylor polynomial of $q_S(x;0)$ through degree $N_S - 1$. The two-endpoint finite part of each monomial $x^{j-N_S}(\log x)^k$, $j < N_S$, is zero, so

$$-B_S(0) = \operatorname{FP}\int_0^\delta P_S(x;0)x^{-N_S}\lambda_S(\log x)\,\mathrm{d}x.$$

Thus the limit is precisely the directly merged finite part of the subset contribution. Summing the finitely many subsets and the outside integral proves (6.10). The proof did not fix an ordering chamber; there are only finitely many choices for the minimum, and the same dominating function applies to all of them. □

Proof of finite polylogarithmic reduction. For distinct nodes, the rational principal part is

$$\frac{P_S(x;\boldsymbol{s})}{D_S(x;\boldsymbol{s})} = \sum_{i\in S}\sum_{k=1}^{d_i}\frac{b_{S,i,k}(\boldsymbol{s})}{(x+s_i)^k}, \tag{6.16}$$

where

$$b_{S,i,k} = \frac{1}{(d_i-k)!}\,\partial_x^{d_i-k}\frac{q_S(x;\boldsymbol{s})}{\prod_{j\in S\setminus\{i\}}(x+s_j)^{d_j}}\bigg|_{x=-s_i}. \tag{6.17}$$

These are finite expressions in the stated tail and weight data. Translate to $y = x + b_S$. All letters $-a_i$, where $a_i = s_i - b_S \ge 0$, lie on the nonpositive real axis. For $a > 0$,

$$\log(y+a) = \log a + G(-a;y), \qquad G(-a;y) = \int_0^y \frac{\mathrm{d}t}{t+a},$$

while $G(0;y) = \log y$ uses its usual tangential normalization. The standard iterated-integral shuffle rule [14] expands each logarithmic product into words of length at most M. A simple pole in (6.16) prepends one letter and therefore gives weight at most $M+1$.

For a pole of order $k \geq 2$, integration by parts gives a boundary term of logarithmic degree at most M and rational integrals of logarithmic degree at most $M-1$. Repeat after partial fractions; induction on M terminates, even if a repeated pole is transferred to another letter. This proves the same weight bound. Lower and upper finite parts are the constant terms of the resulting iterated integrals at 0 and ∞; the real integral in (6.9) fixes their branches without crossing any interior letter.

When all $m_i = 0$, every logarithmic polynomial is constant. The upper primitive has zero constant term, and evaluation at the lower endpoint gives precisely the sums of logarithms and reciprocal powers in the index-zero repeated-pole formula. This proves exact compatibility with the previous convention. □

Remark 6.3 (What regularity has been proved). The result asserts continuity after the full subtraction at arbitrary Stieltjes indices. The residual logarithmic integral need not be analytic in the separations. When all $m_i = 0$, its integrand is analytic and the lower endpoint is an analytic coordinate on each ordering chamber; then the earlier stronger analytic-extension statement is recovered. No differentiability of the higher-index remainder is used to manufacture contact terms.

6.3 One logarithmic index: an explicit dilogarithm kernel

At total Stieltjes index one, the universal counterterms require only ordinary dilogarithms. Define, for $a, b \geq 0$,

$$U(b,a) = \mathrm{FP}\int_0^\infty \frac{\log(y+a)}{y+b}\,\mathrm{d}y.$$

The lower finite part is used only when necessary, and the upper subtraction is $\frac{1}{2}\log^2 R$. The following exact table fixes every branch by real positive arguments:

condition	$U(b,a)$
$a = b = 0$	0
$b = 0 < a$	$\zeta(2) + \frac{1}{2}\log^2 a$
$a = 0 < b$	$-\zeta(2) - \frac{1}{2}\log^2 b$
$a = b > 0$	$-\frac{1}{2}\log^2 a$
$a > b > 0$	$\frac{1}{2}\log^2(a-b) + \zeta(2) - \log(a-b)\log b + \mathrm{Li}_2\left(-\frac{b}{a-b}\right)$
$b > a > 0$	$-\frac{1}{2}\log^2(b-a) - \zeta(2) - \log a \log\left(\frac{b}{b-a}\right) - \mathrm{Li}_2\left(-\frac{a}{b-a}\right)$.

(6.18)

The corresponding index-zero kernel is

$$V(b) = \mathrm{FP}\int_0^\infty \frac{\mathrm{d}y}{y+b} = -\log b \quad (b>0), \qquad V(0) = 0.$$

Proof. If $a > b$, put $d = a - b$. A primitive is

$$\log d \log(y+b) - \mathrm{Li}_2\left(-\frac{y+b}{d}\right).$$

If $b > a$, put $d = b - a$. A primitive is

$$\log(y+a)\log\left(1+\frac{y+a}{d}\right) + \mathrm{Li}_2\left(-\frac{y+a}{d}\right).$$

Differentiation verifies both formulas on the real interval of integration. Evaluate their lower constant terms and use

$$\mathrm{Li}_2(-t)+\mathrm{Li}_2(-1/t)=-\zeta(2)-\tfrac{1}{2}\log^2 t,\qquad t>0,$$

with the correct normalization $-\zeta(2)=-\pi^2/6$, to take the upper constant. The identity follows itself by differentiation and evaluation at $t=1$, where $\mathrm{Li}_2(-1)=-\zeta(2)/2$. Direct logarithmic primitives handle $a=b$ and the endpoints $a=0$ or $b=0$. □

As a normalization check, the two-factor case with shifts $(0,a)$ gives

$$T_{(1,0)}(0,a)=\frac{\frac{1}{2}\log^2 a+\zeta(2)}{a},\qquad T_{(0,1)}(0,a)=\frac{\log^2 a+\zeta(2)}{a}.$$

These are precisely the asymmetric binary collision polynomials already proved in the manuscript. Their unequal leading coefficients are retained by the general construction.

6.4 Three new mixed cubic collision constants

For $0<\epsilon<1/4$ and $j=0,1,2$, set

$$C_j(\epsilon)=\mathrm{FP}\int_{\mathbb{R}/\mathbb{Z}}\gamma_1(\{x+j\epsilon\})\prod_{\substack{0\le k\le 2\\ k\neq j}}\gamma_0(\{x+k\epsilon\})\,\mathrm{d}x,\qquad Q=\mathrm{FP}\int_0^1\gamma_1(x)\gamma_0(x)^2\,\mathrm{d}x. \tag{6.19}$$

The same global constant Q occurs in all three limits, because their merged products agree pointwise before regularization. Put

$$h(x)=\gamma_0(1+x),\qquad k(x)=\gamma_1(1+x),\qquad L=\log\epsilon,\quad \ell=\log 2,\quad B(X)=\zeta(2)+\tfrac{1}{2}X^2.$$

Define

$$\begin{aligned}A_0(L)&=\tfrac{1}{4}L^2-\tfrac{1}{2}\ell L+\tfrac{1}{2}\zeta(2)-\tfrac{1}{4}\ell^2,\\ A_1(L)&=\tfrac{1}{2}L^2-\tfrac{1}{2}\ell L+\tfrac{1}{4}\zeta(2),\qquad A_2(L)=\tfrac{1}{2}L^2.\end{aligned} \tag{6.20}$$

Proposition 6.4 (Exact local counterterms for the three placements)**.** *The counterterms of Definition 6.1 are*

$$T_0(\epsilon)=\frac{A_0(L)}{\epsilon^2}+\frac{h(\epsilon)}{\epsilon}B(L)+\frac{h(-\epsilon)}{2\epsilon}B(L+\ell)+\frac{k(-2\epsilon)}{\epsilon}L, \tag{6.21}$$

$$T_1(\epsilon)=\frac{A_1(L)}{\epsilon^2}+\frac{h(2\epsilon)}{\epsilon}B(L)+\frac{h(\epsilon)}{2\epsilon}L^2+\frac{k(-\epsilon)}{2\epsilon}(L+\ell)+\frac{h(-2\epsilon)}{\epsilon}B(L), \tag{6.22}$$

$$\begin{aligned}T_2(\epsilon)=&\frac{A_2(L)}{\epsilon^2}+\frac{k(\epsilon)}{\epsilon}L+\frac{h(\epsilon)}{2\epsilon}B(L+\ell)+\frac{h(-\epsilon)}{4\epsilon}(L+\ell)^2\\ &+\frac{h(-\epsilon)}{\epsilon}B(L)+\frac{h(-2\epsilon)}{2\epsilon}L^2.\end{aligned} \tag{6.23}$$

In particular $C_j(\epsilon)-T_j(\epsilon)\to Q$ for each placement.

Proof. Only subsets of two and three poles contribute. For the three-pole subset the principal numerator is 1, and

$$\frac{1}{x(x+\epsilon)(x+2\epsilon)}=\frac{1}{2\epsilon^2 x}-\frac{1}{\epsilon^2(x+\epsilon)}+\frac{1}{2\epsilon^2(x+2\epsilon)}.$$

Insert respectively $\log x$, $\log(x+\epsilon)$ and $\log(x+2\epsilon)$, and apply (6.18). The answers are $A_j(L)/\epsilon^2$.

For a pair of poles at shifts $a < b$, with omitted analytic tail q, the principal part is

$$\frac{q(-a)}{(b-a)(x+a)} - \frac{q(-b)}{(b-a)(x+b)}.$$

When the omitted index is one, use V; otherwise use U with the logarithm at the indicated pole. The three pair subsets give exactly the remaining terms of (6.21)–(6.23). Singleton contributions vanish as already proved. The limit is Theorem 6.2. □

For an expression having a power–log expansion through order zero, $\mathrm{CT}_{\epsilon\downarrow 0}$ denotes the coefficient of $\epsilon^0(\log\epsilon)^0$. It discards negative powers of ϵ and nonconstant powers of $\log\epsilon$, including the latter at power ϵ^0.

Corollary 6.5 (The mixed finite anomalies)**.** *With* $\beta = \zeta(2) + \zeta'(2)$,

$$\mathrm{CT}_{\epsilon\downarrow 0}\, C_0(\epsilon) = Q - \tfrac{1}{2}\zeta(2)^2 + \tfrac{1}{4}\zeta(2)\log^2 2, \tag{6.24}$$

$$\mathrm{CT}_{\epsilon\downarrow 0}\, C_1(\epsilon) = Q - \tfrac{1}{2}(\zeta(2) + \zeta'(2))\log 2, \tag{6.25}$$

$$\mathrm{CT}_{\epsilon\downarrow 0}\, C_2(\epsilon) = Q + \tfrac{1}{2}\zeta(2)^2. \tag{6.26}$$

Consequently the global mixed constant cancels in the exact identity

$$\mathrm{CT}_{\epsilon\downarrow 0}(C_2(\epsilon) - C_0(\epsilon)) = \zeta(2)^2 - \tfrac{1}{4}\zeta(2)\log^2 2. \tag{6.27}$$

Proof. The necessary analytic expansions are

$$h(x) = \gamma - \zeta(2)x + O(x^2), \qquad k(x) = \gamma_1 + \beta x + O(x^2).$$

Here $\beta = k'(0)$ follows from (6.5); the derivative of the Pochhammer factor contributes the term $\zeta(2)$. Substitute into (6.21)–(6.23). Every omitted term is $O(\epsilon(1 + |\log\epsilon|)^2) = o(1)$, whereas $C_j - T_j = Q + o(1)$. Taking the constant coefficients proves all three formulas. □

The first placement, with every divergent and finite logarithmic term shown, is

$$\begin{aligned} C_0(\epsilon) = {} & \frac{1}{\epsilon^2}\left\{\frac{L^2}{4} - \frac{\ell L}{2} + \frac{\zeta(2)}{2} - \frac{\ell^2}{4}\right\} \\ & + \frac{1}{\epsilon}\left\{\frac{3\gamma L^2}{4} + \left(\frac{\gamma\ell}{2} + \gamma_1\right)L + \frac{3\gamma\zeta(2)}{2} + \frac{\gamma\ell^2}{4}\right\} \\ & - \frac{\zeta(2)L^2}{4} + \left(\frac{\zeta(2)\ell}{2} - 2\beta\right)L + Q - \frac{\zeta(2)^2}{2} + \frac{\zeta(2)\ell^2}{4} + o(1). \end{aligned} \tag{6.28}$$

The finite constants in (6.24)–(6.26) depend on which factor carries the logarithmic Stieltjes singularity. Retaining only the leading inverse-power divergence would therefore miss the exact answers.

6.5 An ordinary integral for the remaining merged constant

The constant Q in (6.19) is not left without an independent ordinary-integral definition. Local Laurent subtraction gives

$$\begin{aligned} Q = {} & -\frac{1}{4} - 2\gamma - \gamma_1 \\ & + \int_0^1 \left\{ \frac{2\big(h(x) - \gamma + \zeta(2)x\big)\log x}{x^2} + \frac{\big(h(x)^2 - \gamma^2\big)\log x}{x} \right. \\ & \qquad \left. + \frac{k(x) - \gamma_1 - \beta x}{x^2} + \frac{2\big(h(x)k(x) - \gamma\gamma_1\big)}{x} + h(x)^2 k(x) \right\} \mathrm{d}x. \end{aligned} \tag{6.29}$$

Each numerator has the stated vanishing order, so the integral is absolutely convergent. To verify the constant, expand $(\log x/x + k(x))(1/x + h(x))^2$. The pure singular terms have finite integrals

$$\mathrm{FP}\int_0^1 x^{-3}\log x\,\mathrm{d}x = -\frac{1}{4}, \quad \mathrm{FP}\int_0^1 x^{-2}\log x\,\mathrm{d}x = -1, \quad \mathrm{FP}\int_0^1 x^{-2}\,\mathrm{d}x = -1,$$

while the x^{-1} and $x^{-1}\log x$ terms have zero finite integral. This proves (6.29). It supplies independent numerical access to the global constant without converting the proof of the collision identities into a numerical fit.

7 Audit, integration corrections, and further research

7.1 What the audit changes

No new false theorem in the focused established source material was identified. The principal audit finding is a status issue: several questions survive in earlier archives after later incoming reports have answered them. Integration should reconcile the chronology before assigning credit or declaring a question newly resolved.

Earlier question or claim	Proposed integration treatment
Single-resolvent nonpositive powers	The complete answer is already in [2]. The earlier Barnes Question F5 is therefore resolved by that report. The new theorem here concerns independently powered products and their orientation-dependent endpoint poles.
The fourth digamma power	Retain the existing cubic and quartic bridges with credit to [4, 3, 2]. Add the fifth theorem and the all-power reduction as subsequent extensions.
Pure pairs of even harmonic tails	The question in the earlier parity and reflection reports is already answered by [2]. It is not a new contribution of this article. Products with further factors remain a separate problem.
Higher-index collision remainder	Replace the open-status entry for existence of a fully subtracted total limit by Theorem 6.2. State continuity explicitly. Do not upgrade this to analyticity at positive Stieltjes indices or to associativity of arbitrary partial-merger prescriptions.
Canonical power polynomials	Record Corollary 5.2 as elimination of the inherited height-one coefficients η, θ. The additional derivative coordinate χ_4 remains, so the stronger single-coordinate all-power question is still open.
Gaussian S_6 and revised S_8	Preserve their conjectural status. Keep the previously rejected coefficient vector distinct from the revised candidate, and distinguish period equality from membership in a specified formal relation system.

The companion `integration/CLAIM_LEDGER.csv` records theorem labels, source questions, and proposed status changes. The source snapshot manifest gives the inspected revision and hashes of the current incoming archives. The older reports are dependencies and provenance, not copied into this deliverable as if they were new work.

7.2 Exact corrections to tempting extrapolations

Two natural extrapolations are disproved by the calculations here. They should be excluded explicitly in any consolidated statement.

A value along a parameter path is not automatically the coordinate finite part. For opposite resolvents, (2.24) gives the three limits $-2\pi\mathrm{i}, -\pi\mathrm{i}, 0$ for the same specialized function $K - \pi\mathrm{i}$. Its coordinate finite part is $-\pi\mathrm{i}$. The proposed replacement is the full germ (2.19), retaining $E = \sum \epsilon_j \lambda_j$ as well as $\Lambda = \sum \lambda_j$. The phase numerator may be simplified using integrality of each exponent only when that hypothesis actually holds. The theorem itself requires only integrality of the total at the chosen resonance.

Taking the order-zero term of a collision does not commute with merging its factors. The three constants (6.24)–(6.26) differ although their directly merged integrand is the same $\gamma_1\gamma_0^2$. The replacement is to subtract the entire counterterm (6.9) before passing to the limit. Dividing the local cutoff by a shrinking gap would alter this convention and its constants. The terms involving $\gamma_1'(1) = \zeta(2) + \zeta'(2)$ must retain both summands; differentiation of the Pochhammer factor supplies the first one.

These counterexamples correct unsupported extensions. They are not retroactive claims that the earlier single-resolvent or index-zero theorems were false.

Two further proof requirements should accompany the integration. Matching principal parts of ψ^k leaves an entire-function ambiguity; Lemma 4.2 supplies the contour argument that eliminates it. Also, an exact finite harmonic identity is not the same statement as its asymptotic regularization: use (4.17) with finite terminal harmonic sums first, and replace convergent terminal words by their limits only after controlling their tails. These requirements are met in the proofs above.

7.3 Questions opened by the new results

The following projects ask for exact identities or structural theorems. They are questions rather than unproved assertions of reducibility. Each has a concrete starting formula in this article.

Research question 7.1 (The derivative coordinate of weight four)**.** Can

$$\chi_4 = 2(\partial_1 - \partial_2)Z_2(3,1) + \partial_1 Z_2(2,2)$$

be reduced to a smaller specified algebra of ordinary zeta derivatives and height-one Laurent coefficients?

The first task is to differentiate identities valid with genuinely free orders, rather than differentiate a duality known only at discrete integer indices. The free-order stuffle identity used after (5.16) is admissible in this sense. A useful result would be an exact reduction or a formal separating functional for a precisely enumerated family of differentiated relations. Such a formal obstruction would still not prove arithmetic independence.

Research question 7.2 (The sixth power and canonical cancellation matrices)**.** Compute the complete sixth-power reduction and determine which polynomials in ψ eliminate its inherited lower-depth coordinates. Does a uniform family of power polynomials remove all but the single height-one coefficient at every degree?

The compiler of Theorem 4.1 already terminates at every fixed degree and fixes its constants. What remains is to simplify the resulting moving-order coordinates and compute the exact cancellation matrix against lower powers. The coefficient of the top-depth term is already known, namely $-k!$; it is the lower-depth relations that must be understood. The surviving χ_4 in the fifth case is a specific test.

Research question 7.3 (A common Hurwitz shift for all digamma powers)**.** Extend the reduction and the normalized primitive to $\psi(x+a)^k$, with a free common shift a, and determine the finite multiple-Hurwitz or Lerch family required.

At $a > 0$ the unit-interval integral is ordinary; the principal poles of the global meromorphic function have moved to $-a - \mathbb{N}_0$. The contour subtraction and finite summation-by-parts method should be carried out with the shifted lattice, keeping the endpoint normalization explicit. Rational a offers a precise next specialization to colored multiple-zeta data.

Research question 7.4 (Higher mixed logarithmic resolvent products)**.** For arbitrary positive integers r, s, find finite multiple-polylogarithmic expressions for the all-index transforms of $L_z^r L_w^s$. What depth and moving-order derivative families are sufficient?

The two-factor formula (3.5) and the ordinary cubic formula (3.4) are initial cases. Start from the integer convolution constraints of the Fourier coefficients, apply exact partial fractions, and keep the complete local power–log polynomial in the Stieltjes subtraction. A finite representation in a stated class is the target; introducing an unrestricted new generating integral alone would not provide that reduction.

Research question 7.5 (Classical multivariable forms of the Gamma spectrum)**.** For which exponent patterns and disk coordinates does (2.7) collapse to a finite combination of Gauss, Appell, Lauricella, or multiple-polylogarithmic functions?

The example (2.9) proves that a Gauss function suffices for one zero disk coordinate. The general Gamma convolution is already explicit and normally convergent. A further result should produce a smaller function class, a new functional equation, or an arithmetic special-value identity, with all exceptional parameter restrictions handled on the complete meromorphic expression.

Research question 7.6 (Real-axis boundary identities)**.** Determine the exact limits of the completed mixed resolvent transforms as one or more nonreal parameters approach the real axis, including coincident limiting parameters.

The sign ϵ_j then records which boundary value is taken, and interior cotangent poles can appear. A correct statement must inventory their local principal-value and contact contributions as well as the endpoint terms already present in (2.19). This is an exact boundary-value problem, not a request for numerical bounds.

Research question 7.7 (Regularity of the higher-index collision remainder)**.** Find the complete power–log expansion of $C - T$ near total collision, and determine its differentiability and analytic subfamilies.

Theorem 6.2 proves continuity uniformly in the relative gaps. It does not assert a convergent Taylor series. After the Hermite subtraction the only obstacle is the explicit product of translated logarithms. Resolving that remaining integral for one cluster and then for nested clusters would give a controlled higher-order contact calculus, including gap configurations not described by a single scale.

Research question 7.8 (The global mixed cubic constant)**.** Express $Q = \mathrm{FP}\int_0^1 \gamma_1(x)\gamma_0(x)^2\,\mathrm{d}x$ in a specified multiple-Hurwitz or directional Tornheim derivative family, and seek combinations that eliminate it.

Equation (6.29) is an absolutely convergent representative. The difference of collision constants already eliminates Q exactly. For a further arithmetic reduction, begin with a joint completed three-factor Hurwitz generator and differentiate its orders before specialization. A numerical relation for the displayed decimal would only suggest a candidate.

Research question 7.9 (Compatibility of successive partial mergers)**.** Characterize partial-cluster subtraction rules whose iterated mergers agree with the simultaneous full counterterm (6.9). Can a universal local adjustment make the operation associative?

The present result fixes the total limit even for hierarchical gaps. It does not identify the finite constants produced by every possible intermediate prescription. For three and four factors, compare two merger orders symbolically using the explicit dilogarithm kernels, then determine the local difference before proposing a general rule.

Research question 7.10 (Integer-frequency grids and their logarithmic scales)**.** Extend the geometric theorem to $\prod_i \gamma_{m_i}^{(r_i)}(\{p_i x + s_i\})$ with positive integers p_i, including analytic Fourier weights.

The analytic-weight portion at common unit frequency is proved here. For a grid point with ambient displacement t, the singular factor has amplitude $p_i^{-r_i-1} t^{-r_i-1} L_{m_i,r_i}(\log t + \log p_i)$. A complete theorem should inventory all coinciding grid points and transport the cutoff in this ambient coordinate. The logarithms $\log p_i$ are finite contacts that cannot be removed by an informal rescaling.

Research question 7.11 (A common spectral and collision completion)**.** Construct a joint generator with both the complex resolvent-power weight and moving periodic Stieltjes factors, and prove compatible restriction formulas at intersecting spectral and geometric resonances.

The two ingredients are explicit here, but their direct combination has new local products: a weighted collision point can approach an endpoint where the weight is singular. The target is one full subtraction rule that recovers (2.19) and (6.9) in their separate domains, with the order of all restrictions specified.

Research question 7.12 (A new exact relation for the frozen Gaussian targets)**.** Can an identity obtained from a harmonic, Stieltjes, or resolvent transform supply a genuine functional relation for the frozen S_6 or revised S_8 residual outside the previously obstructed formal systems?

The source vectors and branch conventions must remain fixed. A useful candidate should have a demonstrably new action on the existing separating functional, or a proved analytic specialization beyond the enumerated relation family. Neither additional digits nor an uncompleted modular elimination establishes the period equality. No such functional bridge has been proved in this article.

Research question 7.13 (Reusable formalization units)**.** Formalize the full-amplitude completion lemma, the all-power Mittag–Leffler normalization, strict harmonic quasi-shuffles and their asymptotic regularization, and the fixed-contour Hermite remainder.

These units separate the reusable analytic proof from finite symbolic coefficient calculations. The supplied programs give regression examples for the latter. Formal verification of the analytic lemmas would support generated identities at many new indices; importing Python outputs alone should not confer a proof-assistant status.

7.4 Suggested order of work

The most direct next exact calculations are the sixth-power normal form, the first higher mixed logarithmic transform, and a joint Hurwitz representation of the global mixed cubic constant. Their input algebra and all local corrections are now explicit. The collision regularity and merger-compatibility questions are the next structural problems. The Gaussian conjectures require a new functional relation or an exact period certificate beyond the relation systems already analyzed in the source material.

8 Verification, reproducibility, and the evidence boundary

8.1 Exact calculations and independent numerical methods

Every main assertion above is supported by an analytic proof or a finite algebraic derivation. The code checks sensitive coefficients and supplies independent numerical comparisons. Normal convergence, meromorphic continuation, and the hierarchy-uniform collision limit are proved in the article; they are not inferred from numerical agreement.

The harmonic compiler retains exact rational and symbolic coefficients. Its sixty finite checks include the principal-part polynomial, the printed fifth finite harmonic identity, the moving-order finite lattices through degree six, and the separate regularization constants. The classical convergent multiple-zeta identities used in the reduction are stated explicitly in (5.10); they are mathematical inputs, not theorems established by numerical testing.

The two fifth-power evaluators use different analytic representations. One integrates the convergent local Taylor series of $(x\psi(x))^5$ on $(0, 1/8)$ and uses ordinary quadrature on the remainder. The other sums (5.4), using a finite polygamma asymptotic tail represented by Hurwitz-zeta order derivatives. At 70 working decimal digits their observed difference is $3.3937772261 \cdot 10^{-49}$. Thus the recorded agreement is at the 48-decimal-place level; a 70-digit working precision is not itself a 70-digit accuracy certificate.

The opposite-orientation tests verify the residue polynomial exactly for $M = 1, \dots, 9$, and the first logarithmic contact coefficients symbolically. Direct beta quadrature at complex exponents has observed residual $7.37 \cdot 10^{-83}$ at 80 working digits. Four transverse rays and three curved paths recover their distinct uncompleted limits and the common completed value.

The general Fourier convolution is also tested at three distinct nonzero disk coordinates, with two upper orientations and one lower orientation. The exponents are complex, one has negative real part, and the total has real part 1.60. At 80 working digits, the five modes $k = -2, -1, 0, 1, 2$, computed on the rectangle $0 \le n, m \le 160$, differ from direct cotangent-resolvent quadratures by at most $2.20 \cdot 10^{-80}$. A shorter rectangle through 120 gives observed changes below $5.7 \cdot 10^{-63}$. At exponents $(2, 1)$ and zero disk coordinates the spectrum is finite. The exponential moment is checked at four complex values, including a reciprocal-Gamma zero, and the log-Gamma formula (2.17) has observed residual below $9.3 \cdot 10^{-84}$.

For the mixed-log transform, direct quadrature of the Bernoulli polynomial $\zeta(-1, x)$ against the logarithmic product is compared with separately evaluated multiple-polylogarithmic sums at three disk coordinate pairs. One pair is complex and one is $(0, 0)$. The largest observed spectral residual is $6.60 \cdot 10^{-82}$, and the mean comparisons are below $6.4 \cdot 10^{-81}$ at 80 working digits.

There is also a direct singular-integral check of (3.14). The left side uses coordinate subtraction of its actual endpoint polynomial. Independently, η is evaluated by a subtracted positive-real Mellin integral. Both methods give

$$-6.19739612946263280992976519687141974516202856511007559074 75\dots.$$

The observed residual rounds to zero at the 80-digit working precision; the code checks a tolerance of 10^{-70}. This is an observed floating-point agreement, not an interval enclosure.

8.2 Independent geometric-collision checks

The script `verify_collision_kernels.py` compares all cases of the dilogarithm table with independently subtracted ordinary integrals, using 70 working decimal digits. Its eight comparisons have maximum observed absolute residual $1.82 \cdot 10^{-71}$. It also checks the three all-pole coefficients and the three finite anomalies by exact symbolic algebra, treating $\zeta(2), \zeta'(2), \gamma, \gamma_1, \log 2$ as independent symbols. Each symbolic residual is exactly zero.

The separate script `verify_higher_collision.py` evaluates the circle integrals by subtracting their actual local singularities at each of the three points. It does not use the proposed T_j

formulas to define those integrals. It evaluates the directly merged ordinary representative (6.29) independently, obtaining

$$Q = -2.87039237858604022876331895417289849797325401866\ldots.$$

At five gaps from 0.03 down to 0.0003, the quantities $C_j - T_j - Q$ approach zero for every placement. Selected values are

ϵ	$j = 0$	$j = 1$	$j = 2$
0.03	−0.0828460	0.184968	0.608974
0.003	−0.00940356	0.0289980	0.0833617
0.0003	−0.000951212	0.00376258	0.0107961

These are finite-separation remainders, not equality errors expected to vanish at each positive gap. The working precision is 65 decimal digits. A convergent Hurwitz Taylor evaluator for $\gamma_1(1 + x)$ was separately compared with the generalized Stieltjes function at four points, with observed residual below $9 \cdot 10^{-67}$.

The numerical paths above illustrate the limit and check the explicit formulas. Uniformity for arbitrarily hierarchical gaps follows from the fixed-contour and dominated-convergence proof, not from sampling a finite collection of geometric paths.

8.3 Companion files and replay

The ZIP includes the complete modular LaTeX source, a flattened standalone source, the compiled PDF, scripts and raw JSON records, source hashes, and a claim-by-claim integration ledger. The article can be built without the ProveIt checkout or network access. The Python calculations require Python 3, SymPy, and mpmath; the recorded environment uses Python 3.12.14, SymPy 1.14.0, and mpmath 1.3.0.

File in `verification/`	Purpose
`harmonic_power_reduce.py`	General exact finite harmonic compiler; accepts a power on the command line.
`verify_harmonic_powers.py`	Sixty exact finite identities and normal-form comparisons.
`verify_fifth_numeric.py`	Independent Taylor/quadrature and polygamma-series evaluations of Q_5.
`check_mixed_orientation.py`	Exact residue and logarithmic coefficients; beta and path diagnostics.
`verify_multiple_resolvents.py`	Three-parameter complex-power Gamma convolution against direct Fourier quadrature, plus exponential and log-Gamma moment checks.
`check_mixed_log.py`	Ordinary mixed-log spectral transform and dilogarithmic mean.
`verify_special_mixed_log.py`	Independent Mellin and coordinate finite-part check of (3.14).
`verify_collision_kernels.py`	Dilogarithm-kernel quadratures and exact cubic anomaly calculations.
`verify_higher_collision.py`	Direct singularity-subtracted circle quadratures and the merged constant.

The top-level `run_checks.py` explains and runs the replay sequence; the verification README distinguishes exact checks, observed numerical residuals, and finite-separation remainders. The build scripts retain the same article content in modular and standalone forms.

References

[1] ProveIt Contributors, *Polylogarithms and their Arithmetic Bridges*, four-volume manuscript and source status records, repository revision `9035bb0eaaf70f3398711b489733093dcc52e946`. https://github.com/VladimirReshetnikov/ProveIt/tree/9035bb0eaaf70f3398711b489733093dcc52e946/Analysis/Polylogarithms/docs/manuscript. The companion incoming directory was inspected at the same revision.

[2] ProveIt incoming report, *Quartic Tails and Complex Resolvents*, 11 October 2026. Archive `ProveIt_Quartic_Tails_Resolvents_2026-10-11.zip`, at the revision in [1]. Relevant members: `sections/quartic_bridge.tex`, `sections/complex_powers.tex`, and `sections/research_questions.tex`, Questions 1–2 and 10–11. https://github.com/VladimirReshetnikov/ProveIt/blob/9035bb0eaaf70f3398711b489733093dcc52e946/docs/incoming/ProveIt_Quartic_Tails_Resolvents_2026-10-11.zip.

[3] ProveIt incoming report, *Barnes Resolvents and Higher Harmonic Identities*, 11 October 2026. Archive `ProveIt_Barnes_Resolvents_and_Harmonic_Identities_2026-10-11.zip`, at the revision in [1]. Relevant members: `sections/05_quartic_harmonic.tex` and `sections/07_audit_research.tex`, Questions F3 and F5. https://github.com/VladimirReshetnikov/ProveIt/blob/9035bb0eaaf70f3398711b489733093dcc52e946/docs/incoming/ProveIt_Barnes_Resolvents_and_Harmonic_Identities_2026-10-11.zip.

[4] ProveIt incoming report, *Cubic Tornheim Derivatives and Harmonic Stieltjes Constants*, 11 October 2026. Archive `ProveIt_Cubic_Harmonic_Mixed_Orders_2026-10-11.zip`, at the revision in [1]. Relevant member: `sections/02_cubic_harmonic.tex`. https://github.com/VladimirReshetnikov/ProveIt/blob/9035bb0eaaf70f3398711b489733093dcc52e946/docs/incoming/ProveIt_Cubic_Harmonic_Mixed_Orders_2026-10-11.zip.

[5] ProveIt incoming report, *Higher Reflection Identities and Resonant Zeta Calculus*, 11 October 2026. Archive `ProveIt_Higher_Reflection_Identities_2026-10-11.zip`, at the revision in [1]. Relevant members: `sections/04_collision.tex`, Theorems `gc:main` and `gc:derivative`, and `sections/06_audit_research.tex`, the higher Stieltjes geometric collision and integer-frequency/weight questions. https://github.com/VladimirReshetnikov/ProveIt/blob/9035bb0eaaf70f3398711b489733093dcc52e946/docs/incoming/ProveIt_Higher_Reflection_Identities_2026-10-11.zip.

[6] ProveIt incoming report, *Exact Resonant Identities Beyond the Diagonal*, 11 October 2026. Archive `ProveIt_Exact_Resonant_Identities_2026-10-11.zip`, at the revision in [1]. Relevant members: source audit, research-status section, and the specified Gaussian/Cayley relation certificates. https://github.com/VladimirReshetnikov/ProveIt/blob/9035bb0eaaf70f3398711b489733093dcc52e946/docs/incoming/ProveIt_Exact_Resonant_Identities_2026-10-11.zip.

[7] NIST Digital Library of Mathematical Functions, §5.12, *Beta Function*, especially Eq. 5.12.6, the Fourier beta integral. https://dlmf.nist.gov/5.12.E6.

[8] NIST Digital Library of Mathematical Functions, §25.11, *Hurwitz Zeta Function*, including the recurrence, Fourier series, argument derivatives, and the derivative at order zero. https://dlmf.nist.gov/25.11.

[9] NIST Digital Library of Mathematical Functions, §5.7, *Series Expansions*, Gamma and digamma series. https://dlmf.nist.gov/5.7.

[10] NIST Digital Library of Mathematical Functions, §5.15, *Polygamma Functions*, especially the positive-integer values and recurrence. https://dlmf.nist.gov/5.15.

[11] NIST Digital Library of Mathematical Functions, §§5.5, 5.7, 5.11, 25.2, and 25.16, digamma recurrence and reflection, local and asymptotic expansions, the Stieltjes convention, and Euler sums. https://dlmf.nist.gov/5.5, https://dlmf.nist.gov/5.11, https://dlmf.nist.gov/25.16.

[12] M. E. Hoffman, *Algebraic aspects of multiple zeta values*, in *Zeta Functions, Topology and Quantum Physics*, T. Aoki et al. (eds.), Springer, 2005, pp. 51–74. Preprint arXiv:math/0309425v2, 2004. https://arxiv.org/abs/math/0309425.

[13] I. Mező, *Nonlinear Euler sums*, Pacific Journal of Mathematics **272** (2014), no. 1, 201–226. DOI: 10.2140/pjm.2014.272.201. https://msp.org/pjm/2014/272-1/pjm-v272-n1-p09-s.pdf.

[14] A. B. Goncharov, *Multiple polylogarithms and mixed Tate motives*, arXiv:math/0103059v4, 2001. https://arxiv.org/abs/math/0103059.